

Green subsidies with demand distortions

On-line Appendix

Susanna B. Berkouwer[§]

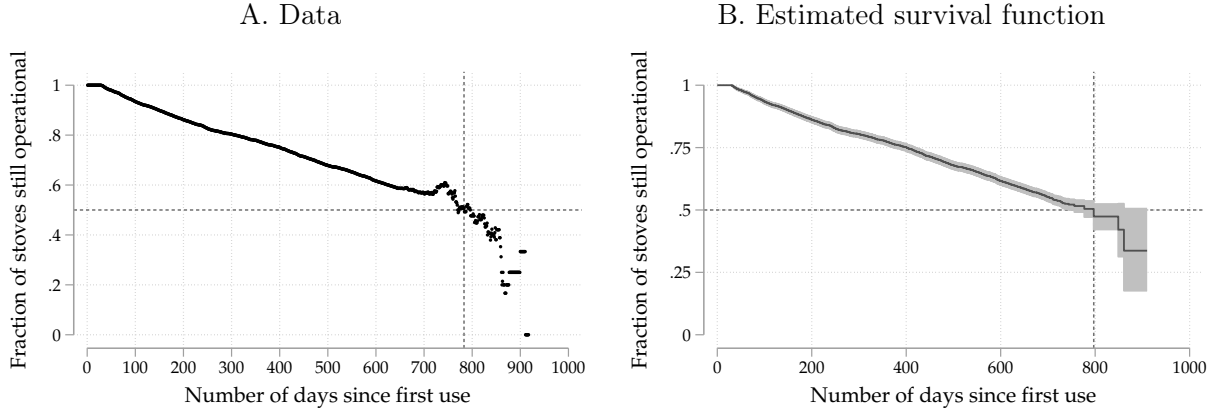
Joshua T. Dean[†]

October 9, 2026

[§]The Wharton School, University of Pennsylvania and NBER; sberkou@wharton.upenn.edu. [†]The Booth School of Business, University of Chicago; joshua.dean@chicagobooth.edu. The main paper is [available here](#).

III Online Appendix Figures and Tables

Figure B1: Stove lifespan estimation



Notes: Stove usage rates for the 3,797 stoves that had been sold on or before December 31, 2024. The time variable is the number of days between first usage and last usage. An adverse event is the last observed usage if that usage happened more than three months before the end of our dataset. We are agnostic as to the cause of the adverse event: it likely includes breakage as well as non-payment with stove lock-out. We drop the 2% of stoves with fewer than 24 hours of usage as well as the 3% of stoves with usage between 1 and 29 days, because faulty stoves identified within this period are replaced for free by the manufacturer. Median lifespans are marked by the dotted lines. Panel A shows the empirical distribution of survival. Panel B shows the survival curve estimated using Cox proportional hazards, according to which the induction stove sold by the manufacturer has an expected lifespan of 2.2 years.

Table B1: Attrition by treatment assignment and adoption status

	Control Mean	Attrited (Visit 3)
Credit treatment (=1)	0.030	-0.010 (0.007)
BDM price < 30 USD	0.027	-0.005 (0.007)
Electricity treatment > 0	0.021	0.006 (0.007)
Bought induction stove (=1)	0.031	-0.022*** (0.006)
Joint F-test p-value (treatments only)		0.37
Sample Mean		0.02

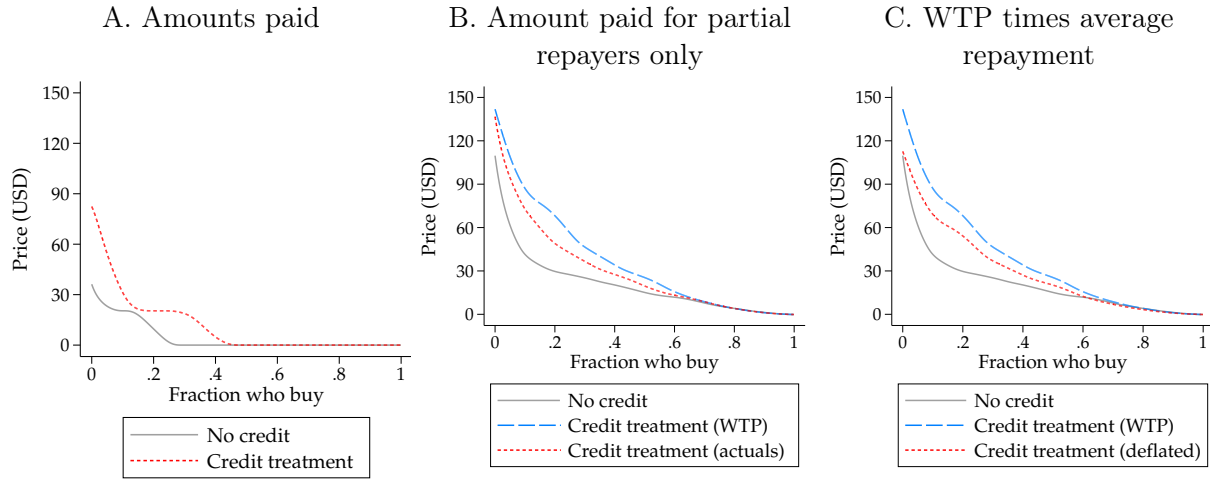
Notes: 2,134 respondents completed the main survey and 2,081 completed the endline survey. The table reports predictors of attrition (defined as not completing an endline visit) among the sample of respondents who completed the main survey. Each row is a regression. The joint F -test is from a single regression of attrition on the three randomized treatment indicators and excludes adoption. Heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B2: Attrition: Participant availability

Reason	Frequency
Completed survey	2081
Unavailable	8
Withdrew from study	8
Relocated outside survey team reach	36
Hospitalized	1
Total	2134

Notes: Completion of the endline survey among the 2,134 respondents who completed the main survey. Participants were classified as relocated outside survey team reach if they had moved outside the survey area or outside Kenya either permanently or for at least three months and thus could not be contacted for follow-up.

Figure B2: Willingness-to-pay adjusting for repayment rates



Notes: Demand curves that adjust for observed repayment rates. Panel A presents the actual amounts paid by the credit group, counting non-adopters as zeros. Panel B replaces willingness-to-pay (WTP) with the actual amount paid only for buyers with credit who did not pay 100% of the loan. Panel C deflates WTP across the distribution by the average default rate. Columns 2, 3, and 4 of Online Appendix Table B7 present regression estimates of the impact of access to credit using these three definitions, respectively.

Table B3: Balance on main treatment variables

	Sample Mean	Credit Treatment	Low BDM Price (<30)	Electricity Treatment	N
	(1)	(2)	(3)	(4)	
Stratification variable: Charcoal spending in 50th–75th percentile (=1)	0.35 [0.48]	0.01 (0.02)	-0.00 (0.02)	0.00 (0.02)	2134
Stratification variable: Charcoal spending above 75th percentile (=1)	0.14 [0.35]	0.01 (0.01)	0.01 (0.02)	-0.01 (0.02)	2134
Stratification variable: Uses LPG/gas/ethanol for cooking (=1)	0.58 [0.49]	0.00 (0.02)	0.00 (0.02)	0.00 (0.02)	2134
Stratification variable: Electricity top-up spending above median (=1)	0.54 [0.50]	0.01 (0.02)	-0.01 (0.02)	0.01 (0.02)	2134
Household size	4.93 [1.57]	-0.01 (0.07)	0.07 (0.07)	0.03 (0.07)	2134
Age	35.82 [10.11]	0.05 (0.44)	-0.26 (0.44)	-0.19 (0.47)	2134
Male respondent	0.10 [0.30]	0.00 (0.01)	0.02 (0.01)	0.01 (0.01)	2134
Charcoal cooking time (minutes/day)	185.36 [68.41]	-2.44 (2.97)	-2.98 (2.96)	2.97 (3.15)	2134
LPG/ethanol spending (USD/month)	4.93 [4.66]	-0.01 (0.20)	-0.06 (0.20)	-0.02 (0.22)	2128
Respondent income (USD/month)	100.08 [105.65]	2.07 (4.58)	0.87 (4.59)	-2.96 (4.83)	2127
Household income (USD/month)	243.16 [182.01]	13.97 (8.51)	13.19 (8.61)	-10.14 (9.22)	1815
Blackout hours (last 30 days)	13.06 [21.02]	1.11 (0.91)	0.01 (0.91)	-0.47 (0.97)	2134
Voltage fluctuation hours (last 30 days)	1.75 [6.07]	-0.44* (0.26)	-0.12 (0.26)	-0.05 (0.28)	2134
Joint F-test p-value, non-stratification variables		0.30	0.44	0.60	

Notes: Test for balance of baseline variables. We only include participants who completed the main survey because they are the only participants whom we assigned to a treatment group. Each cell in columns 2–4 is a separate regression of the row variable on the column treatment indicator; heteroskedasticity-robust standard errors in parentheses. Continuous baseline variables are winsorized at the 99th percentile. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B4: Impact of induction stove purchase using only price as the exogenous instrument

A. Energy usage (monthly expenditures, USD)

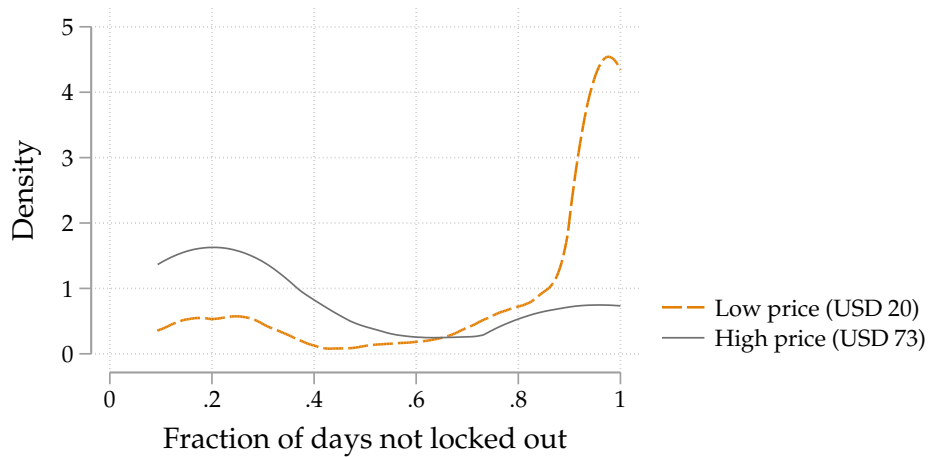
	Charcoal	LPG	Wood	Electricity		Total
	(1)	(2)	(3)	(4)	(5)	(6)
Bought induction stove (=1)	-10.1*** (0.8)	-0.9 (0.6)	-0.3 (0.2)	2.3*** (0.4)	2.1*** (0.7)	-9.0*** (1.0)
Observations	2081	2081	2081	2081	703	2081
Control Mean	15.1	5.5	0.7	3.5	3.7	24.9
F-Stat	714	714	714	714	222	714

B. Emissions (tons of CO₂e per year)

	Charcoal	LPG	Electricity	Total
	(1)	(2)	(3)	(4)
Bought induction stove (=1)	-2.52*** (0.20)	-0.03 (0.02)	0.02*** (0.00)	-2.53*** (0.20)
Observations	2081	2081	2081	2081
Control Mean	3.77	0.17	0.04	3.99
F-Stat	713.78	713.78	713.78	713.78

Notes: The same regressions as in Table 2, but only the randomized treatment price is used as the exogenous instrument, and regressions control for WTP. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure B3: Fraction of days on track by randomly assigned price



Notes: Distribution of the fraction of days a respondent had access to the induction stove, by whether they paid the (randomly assigned) high price or low price. Respondents are 'locked out' when they are behind on their payments.

Table B5: Heterogeneity in energy savings: Socioeconomic predictors

Baseline characteristic	<i>n</i>	Share = 1	Effect of adoption on total energy spending (US\$/month)		Difference	<i>p</i>
			Without the characteristic	With the characteristic		
Age above median	2081	0.48	-7.98 (1.32)	-8.62 (1.23)	-0.64 (1.80)	0.721
Charcoal spending above median	2081	0.49	-9.09 (1.06)	-7.40 (1.53)	1.69 (1.86)	0.364
Owns 2+ charcoal stoves	2081	0.07	-8.27 (0.93)	-12.22 (3.78)	-3.95 (3.90)	0.311
Owns at least one LPG stove	2081	0.57	-10.70 (1.51)	-6.76 (1.13)	3.94** (1.88)	0.036
Ever cooks with firewood	2081	0.14	-8.08 (1.01)	-9.35 (2.08)	-1.27 (2.31)	0.583
Electricity spend above median	2081	0.30	-8.33 (1.11)	-8.24 (1.58)	0.09 (1.93)	0.963

Notes: Each row is a separate instrumental-variables regression of total monthly energy spending (charcoal, LPG, firewood, and electricity, the last inclusive of the experimental subsidy) on adoption of the induction stove, adoption interacted with the binary baseline characteristic named in the row, and the characteristic itself. Adoption and its interaction are both instrumented, by the randomly assigned BDM price, the credit treatment, and their interaction, and by those three interacted with the characteristic. Controls are the stratification variables and the remaining randomized treatment groups. The pooled effect is -US\$8.1 against a complier control mean of US\$24.89 per month. Age is split at the sample median of 35, monthly charcoal spending at US\$16.17 and baseline electricity spending at US\$3.08 per month; electricity spending is bunched at its median, which is why the high group is 30% rather than 50% of households. The two-charcoal-stove and firewood subgroups are small, at 7% and 14% of households. Heteroskedasticity-robust standard errors in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B6: Heterogeneity in energy savings: group average treatment effects

	Effect of adoption on total energy spending (US\$/month)
Average treatment effect	-8.1 (0.9)
Decile 1	-13.3 [-21.5, -5.1]
Decile 2	-11.7 [-20.0, -3.4]
Decile 3	-10.7 [-19.0, -2.4]
Decile 4	-10.0 [-18.2, -1.6]
Decile 5	-9.1 [-17.5, -0.7]
Decile 6	-8.3 [-16.5, -0.0]
Decile 7	-7.5 [-15.7, 0.7]
Decile 8	-6.6 [-14.8, 1.7]
Decile 9	-5.8 [-14.0, 2.5]
Decile 10	-4.1 [-12.3, 4.3]

Notes: The first row reports the pooled instrumental-variables estimate of the effect of induction stove adoption on total monthly energy spending (charcoal, LPG, firewood, and electricity), identical to column 7 of [Table 2](#). The remaining rows report sorted group average treatment effects following Chernozhukov et al. (2025). In each of 100 sample splits, LASSO predictions of the treatment effect are formed on an auxiliary half of the sample, participants in the other half are grouped into deciles of the predicted effect, and the decile-specific effects are estimated jointly in a weighted regression that interacts adoption, demeaned by the known assignment probability, with the decile indicators. Reported estimates are medians across the splits, with deciles ordered by the estimated effect; brackets contain the median 95% confidence bands, Bonferroni-adjusted across the ten groups.

Table B7: Robustness of credit treatment effects to default

	(1) WTP	(2) Amount paid	(3) Amount paid (partial repayers)	(4) Deflated WTP
Credit	14.7*** (1.9)	6.4*** (1.0)	8.5*** (1.7)	7.4*** (1.6)
Observations	2134	2134	2134	2134
Control Mean	19.5	4.3	19.5	19.5

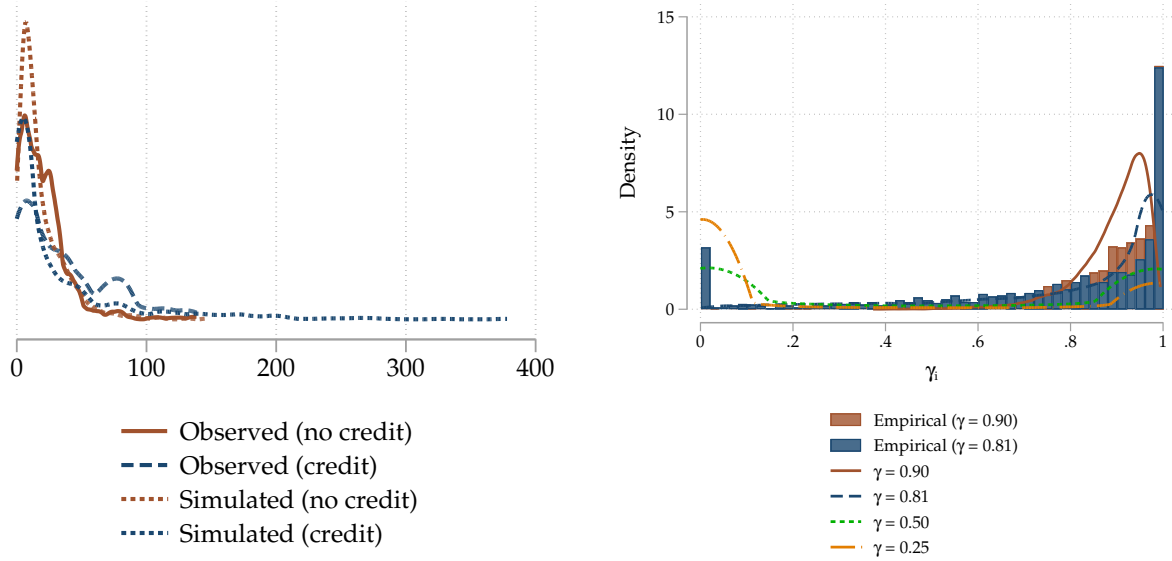
Notes: Robustness of the effect of access to credit on willingness-to-pay (WTP), adjusting for observed repayment rates. Column 1 replicates the main effect of credit on WTP. Column 2 uses the amount of money paid, counting those who did not adopt as zero. Column 3 replaces WTP with the amount paid for those who paid less than 100%. Column 4 deflates WTP in the credit treatment by the average rate of default. Standard errors clustered by respondent in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B8: Aggregate impact of demand distortions on purchase subsidy efficacy, with bootstrapped confidence intervals

	Less distorted ($\hat{\gamma}_1 = 0.81$) (1)	More distorted ($\hat{\gamma}_0 = 0.9$) (2)	Advantage of subsidies in a distorted market (3)
<i>Subsidy cost per tCO₂e (Eq. 12)</i>	US\$20	US\$12	−41% [−51%, −28%]
<i>Observable welfare gain per subsidy dollar (Eq. 13)</i>			
Remaining distortion γ_1 is 100% allocation	US\$7.0	US\$12.1	+73% [+37%, +115%]
Remaining distortion γ_1 is 100% resource	US\$7.5	US\$10.2	+35% [+5.5%, +69%]

Notes: This table replicates Panel B of [Table 6](#) and adds 95% confidence intervals to Column (3), in brackets. Each comparison in that column is a ratio of quantities that are themselves estimated: the shares of participants willing to pay each of the two prices and the mean WTP in each credit arm, the pooled fuel-savings estimate from [Table 2](#), and the marginal-adopter regressions of [Table 4](#). To account for the joint sampling error of these inputs, we draw 500 bootstrap samples of respondents with replacement, stratified by credit arm so that each sample preserves the randomized arm sizes, and recompute every input and every row of the table within each sample. The intervals are the 2.5th and 97.5th percentiles of the resulting distribution of each Column (3) statistic; because these statistics are ratios, the intervals are not symmetric around the point estimates. Resampling respondents rather than regression residuals preserves the covariance between the two channels, since the same households enter both the demand curves and the marginal-adopter regressions.

Figure B4: Empirical and simulated WTP and gamma distributions
A. Willingness-to-pay B. Demand distortion (γ_i)



Notes: Panel A: Simulated and observed distributions of WTP by access to credit. Panel B: The histograms show the empirical distributions of the γ_i obtained by using LASSO to estimate heterogeneous treatment effects, projecting the resulting estimates onto respondent characteristics to predict each respondent's savings, and setting $1 - \gamma_i$ equal to WTP divided by predicted savings. The four lines show $\gamma_i = \text{logistic}(\mu_c + \sigma_c \eta_i)$, $\eta_i \sim N(0, 1)$, with dispersion $\sigma_c = \log(1 + \exp(\Omega_4 + \Omega_5(1 - \gamma_c)))$ for the estimated values of Ω_4 and Ω_5 across four different values of γ_c . Even though GMM estimates five parameters to match seven moments simultaneously, the distributions appear to capture reasonably well the estimated underlying distribution of γ_i .

Table B9: Simulated estimates with standard errors and confidence intervals

A. Estimation

Moment	RCT: Pay in cash ($\gamma = 0.9$)			RCT: Pay with credit ($\gamma = 0.81$)		
	Empirical	Simulated	Deviation	Empirical	Simulated	Deviation
Correlation (ν)	0.80	0.80	-1%	0.80	0.80	-1%
<i>s.e.</i>	(0.010)	(0.001)		(0.010)	(0.001)	
Marginal externality ($\bar{\Phi}$)	5.73	5.81	1%	4.82	4.91	2%
<i>s.e.</i>	(0.360)	(0.019)		(0.371)	(0.020)	
Extensive margin elasticity ($\mathcal{E}_D$)	-2.33	-2.21	6%	-0.88	-0.76	13%
<i>s.e.</i>	(0.133)	(0.025)		(0.053)	(0.009)	
USD per marginal adoption	67.49	67.43	0%	95.47	96.83	1%
<i>s.e.</i>	(0.739)	(0.112)		(2.891)	(0.533)	

B. Counterfactual simulations: Varying the distortion γ

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Assumed demand distortion (γ)	0.95	0.9	0.81	0.75	0.5	0.25	0
Marginal externality ($\bar{\Phi}$)	7.9 [7.3, 13.8]	5.8 [5.5, 9.4]	4.9 [4.7, 7.7]	4.6 [4.5, 6.8]	3.8 [3.7, 4.8]	3.3 [3.2, 4.0]	2.9 [2.8, 3.6]
Extensive margin elasticity ($\mathcal{E}_D$)	-8.2 [-9.0, -2.0]	-2.2 [-2.4, -1.1]	-0.8 [-0.8, -0.6]	-0.5 [-0.5, -0.5]	-0.2 [-0.3, -0.2]	-0.1 [-0.2, -0.1]	-0.1 [-0.2, -0.1]
USD per marginal adoption	64 [64, 69]	67 [66, 82]	97 [94, 110]	125 [123, 130]	235 [185, 247]	321 [210, 349]	397 [225, 449]
<i>Subsidy cost per tCO₂e (in USD)</i>							
Observed		12 [11, 14]	20 [17, 23]				
Simulated	8.0 [5.0, 8.8]	12 [9, 12]	20 [14, 21]	27 [19, 28]	61 [38, 63]	96 [51, 101]	137 [62, 154]

Notes: [Table 7](#) with uncertainty attached. The point estimates are identical: this table does not re-estimate the model; it adds standard errors and intervals to the estimates reported there. Sampling uncertainty in the moments is obtained by resampling respondents and re-running each estimating equation; simulation uncertainty is the spread across simulation seeds at the estimated parameters. In Panel A, the standard error below each empirical moment is the resampling standard error and the one below each simulated moment is the simulation standard error. In Panel B, brackets are 95% intervals. For the simulated rows they are percentiles across 300 parameter vectors drawn from the estimated sampling distribution, with the counterfactual recomputed for each draw; for the two observed rows they come from the moment bootstrap directly. The interval on the subsidy cost per marginal adoption is degenerate in column 1 because at that level of distortion no simulated agent has a willingness-to-pay above the high price $p - s_l = \text{US\$73}$, so every marginal adopter is induced by the full subsidy and the cost equals $s_h = \text{US\$64}$ in every draw. The estimated parameters are $\hat{\Omega}_1 = 1.9$ (0.202), $\hat{\Omega}_2 = 1$ (0.019), $\hat{\Omega}_3 = 0.6$ (0.008), $\hat{\Omega}_4 = -2.3$ (0.153), and $\hat{\Omega}_5 = 24$ (0.77), all significant at the 1% level. The model is overidentified with 2 degrees of freedom; the Hansen J statistic is 22.8 ($p\text{-val} < 0.001$), so the overidentifying restrictions are rejected. The seven moments cannot be matched exactly by five parameters, which Panel A shows directly: the extensive margin elasticity among respondents paying with credit is the moment the model fits least well.

Table B10: Simulated estimates varying both the demand distortion and the correlation between private and social benefits

	Assumed demand distortion (γ)	Assumed correlation (ν)	Average marginal externality ($\bar{\Phi}$)	Extensive margin elasticity ($\mathcal{E}_D$)	Subsidy cost per marginal adoption (US\$)	Subsidy cost per tCO _{2e} (US\$)
(1) <i>Counterfactual</i>	.9	-.8	3.7	-2.3	67.2	18.1
(2) <i>Counterfactual</i>	.9	-.6	4	-2.3	67.2	16.8
(3) <i>Counterfactual</i>	.9	-.4	4.2	-2.3	67.2	15.9
(4) <i>Counterfactual</i>	.9	-.2	4.5	-2.3	67.2	15
(5) <i>Counterfactual</i>	.9	0	4.8	-2.3	67.2	14
(6) <i>Counterfactual</i>	.9	.2	5.1	-2.3	67.2	13.2
(7) <i>Counterfactual</i>	.9	.4	5.3	-2.3	67.2	12.6
(8) <i>Counterfactual</i>	.9	.6	5.6	-2.3	67.2	12.1
(9) RCT: Pay in cash	.9	.8	5.8	-2.3	67.2	11.5
(10) <i>Counterfactual</i>	.9	1	6.6	-2.3	67.2	10.2
(11) <i>Counterfactual</i>	.81	-.8	4.6	-.8	96.4	21.1
(12) <i>Counterfactual</i>	.81	-.6	4.6	-.8	96.4	20.8
(13) <i>Counterfactual</i>	.81	-.4	4.7	-.8	96.4	20.6
(14) <i>Counterfactual</i>	.81	-.2	4.7	-.8	96.4	20.4
(15) <i>Counterfactual</i>	.81	0	4.8	-.8	96.4	20.1
(16) <i>Counterfactual</i>	.81	.2	4.8	-.8	96.4	19.9
(17) <i>Counterfactual</i>	.81	.4	4.9	-.8	96.4	19.8
(18) <i>Counterfactual</i>	.81	.6	4.9	-.8	96.4	19.7
(19) RCT: Pay with loan	.81	.8	4.9	-.8	96.4	19.5
(20) <i>Counterfactual</i>	.81	1	5.1	-.8	96.4	19.1
(21) <i>Counterfactual</i>	.5	-.8	6	-.2	235.4	39.1
(22) <i>Counterfactual</i>	.5	-.6	5.7	-.2	235.4	41.3
(23) <i>Counterfactual</i>	.5	-.4	5.4	-.2	235.4	43.3
(24) <i>Counterfactual</i>	.5	-.2	5.2	-.2	235.4	45.6
(25) <i>Counterfactual</i>	.5	0	4.8	-.2	235.4	48.9
(26) <i>Counterfactual</i>	.5	.2	4.5	-.2	235.4	52.1
(27) <i>Counterfactual</i>	.5	.4	4.3	-.2	235.4	54.8
(28) <i>Counterfactual</i>	.5	.6	4.1	-.2	235.4	57.7
(29) <i>Counterfactual</i>	.5	.8	3.8	-.2	235.4	61.3
(30) <i>Counterfactual</i>	.5	1	3.1	-.2	235.4	75.6
(31) <i>Counterfactual</i>	0	-.8	7.4	-.1	396.6	53.4
(32) <i>Counterfactual</i>	0	-.6	6.7	-.1	396.6	58.9
(33) <i>Counterfactual</i>	0	-.4	6.2	-.1	396.6	64.3
(34) <i>Counterfactual</i>	0	-.2	5.6	-.1	396.6	71.2
(35) <i>Counterfactual</i>	0	0	4.8	-.1	396.6	82.2
(36) <i>Counterfactual</i>	0	.2	4.2	-.1	396.6	93.5
(37) <i>Counterfactual</i>	0	.4	3.8	-.1	396.6	104.8
(38) <i>Counterfactual</i>	0	.6	3.4	-.1	396.6	118
(39) <i>Counterfactual</i>	0	.8	2.9	-.1	396.6	138
(40) <i>Counterfactual</i>	0	1	1.4	-.1	396.6	281.7

Notes: Additional counterfactual estimates using alternative values of demand distortion (γ) and correlation between private and social benefits (ν).

Table B11: Simulated estimates from observed and counterfactual parameters: Gamma-distributed private benefits

A. Estimation						
Moment	RCT: Pay in cash ($\gamma = 0.9$)			RCT: Pay with credit ($\gamma = 0.81$)		
	Empirical	Simulated	Deviation	Empirical	Simulated	Deviation
Correlation (ν)	0.8	0.8	0%	0.8	0.8	0%
Marginal externality ($\bar{\Phi}$)	5.7	5.8	1%	4.8	4.9	1%
Extensive margin elasticity ($\mathcal{E}_D$)	-2.3	-2.2	4%	-0.9	-0.8	13%
USD per marginal adoption	67.5	67.1	-1%	95.5	97.8	2%

B. Counterfactual simulations: Varying the distortion γ							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Assumed demand distortion (γ)	0.95	0.9	0.81	0.75	0.5	0.25	0
Marginal externality ($\bar{\Phi}$)	8.4	5.8	4.9	4.5	3.8	3.3	2.9
Extensive margin elasticity ($\mathcal{E}_D$)	-7.1	-2.2	-0.8	-0.5	-0.2	-0.1	-0.1
USD per marginal adoption	64	67	98	126	241	334	423
<i>Subsidy cost per tCO₂e (in USD)</i>							
Observed		12	20				
Simulated	8	12	20	28	63	101	148

Notes: Robustness check of Table 7 with an alternative distribution of private benefits. Private benefits $v_i/\hat{v}$ and the independent draw ε_i follow a Gamma distribution with shape Ω_1 and scale Ω_2 in place of the Weibull distribution of the main specification. The distortion γ_i is logit-normal as in the main specification, with mean equal to the γ in the column heading and dispersion $\sigma = \log(1 + \exp(\Omega_4 + \Omega_5(1 - \gamma)))$. All five parameters are re-estimated by the same two-step GMM. Panel A reports the moment fit, which can be compared with Table 7. Panel B shows that the distributional assumption does not meaningfully affect the qualitative results from the counterfactual simulations.

Table B12: Simulated estimates from observed and counterfactual parameters: logit- t demand distortion

A. Estimation						
Moment	RCT: Pay in cash ($\gamma = 0.9$)			RCT: Pay with credit ($\gamma = 0.81$)		
	Empirical	Simulated	Deviation	Empirical	Simulated	Deviation
Correlation (ν)	0.8	0.8	-1%	0.8	0.8	-1%
Marginal externality ($\bar{\Phi}$)	5.7	5.8	1%	4.8	4.9	1%
Extensive margin elasticity ($\mathcal{E}_D$)	-2.3	-2.7	-15%	-0.9	-0.8	4%
USD per marginal adoption	67.5	65.8	-3%	95.5	92.1	-4%

B. Counterfactual simulations: Varying the distortion γ							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Assumed demand distortion (γ)	0.95	0.9	0.81	0.75	0.5	0.25	0
Marginal externality ($\bar{\Phi}$)	8.2	5.8	4.9	4.5	3.8	3.2	2.8
Extensive margin elasticity ($\mathcal{E}_D$)	-7	-2.7	-0.8	-0.5	-0.2	-0.1	-0.1
USD per marginal adoption	64	66	92	120	226	317	395
<i>Subsidy cost per tCO₂e (in USD)</i>							
Observed		12	20				
Simulated	8	11	19	26	60	98	140

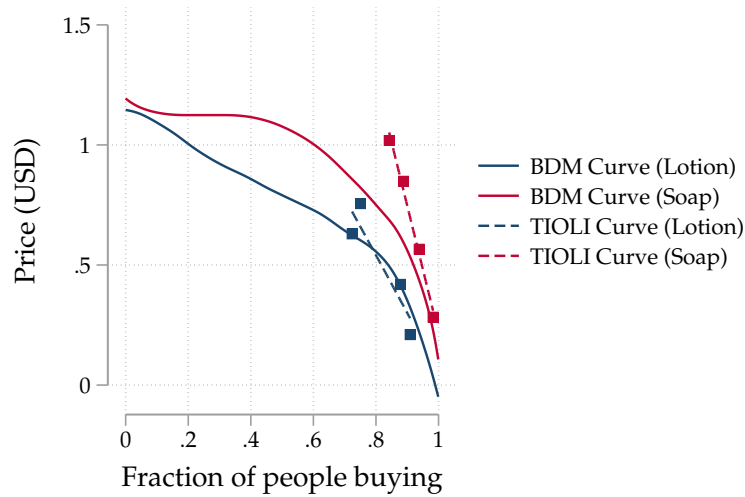
Notes: Robustness check of Table 7 with an alternative distributional assumption: estimating the model with a heavy-tailed logit- t distribution such that $\gamma_i = \text{logistic}(\mu_c + \sigma_c t_i)$, $t_i \sim t_5$ rescaled to $\text{Var}(t_i) = 1$ with $\sigma_c = \log(1 + \exp(\Omega_4 + \Omega_5(1 - \gamma_c)))$. Panel B shows that the distributional assumption does not meaningfully affect the qualitative results from the counterfactual simulations.

Table B13: Simulated marginal value of public funds and economic cost per ton from observed and counterfactual parameters

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Assumed demand distortion (γ)	0.95	0.9	0.81	0.75	0.5	0.25	0
Marginal externality ($\bar{\Phi}$)	7.9	5.8	4.9	4.6	3.8	3.3	2.9
Extensive margin elasticity ($\mathcal{E}_D$)	-8.2	-2.2	-0.8	-0.5	-0.2	-0.1	-0.1
USD per marginal adoption	64	67	97	125	235	321	397
<i>Resource cost per tCO₂e (in USD)</i>							
Observed		-25	-19				
Simulated	-36	-25	-19	-17	-7	1	12
<i>Observable welfare gain per subsidy dollar (in USD)</i>							
Observed		12	7				
Simulated	19	12	7	5.1	2.1	1.2	0.8
<i>Marginal Value of Public Funds (in USD)</i>							
Observed		13	8				
Simulated	20	13	8	6	3	2	2

Notes: Additional versions of Panel B of [Table 7](#). Because the simulations assume $r = 0$, the observable welfare gain and MVPF rows correspond to the case of a pure allocation distortion. The economic cost of abatement is defined as the economic cost per adoption divided by the abatement per adoption: $\frac{p-\hat{v}}{\hat{\Phi}/\text{SCC}}$, where $\hat{v}$ and $\hat{\Phi}$ are the average private fuel savings and the SCC-valued externality per marginal adoption, so that $\hat{\Phi}/\text{SCC}$ is tons abated per adoption. Since fuel savings for the distorted groups exceed the US\$84 production cost of the stove, the economic cost per ton is *negative*: each cookstove *saves* economic resources in this context. Abating one ton of CO₂e thus *reduces* total economic costs by US\$25 at $r = 0$ (with the distortion's resource cost this ranges from US\$0 to US\$23; [Section 5.1](#)). The marginal value of public funds (MVPF) follows Hendren and Sprung-Keyser ([2020](#)). Since willingness-to-pay is distorted and thus cannot be taken as a measure of value, we calculate this directly as the sum of the cash transfer (valued at US\$1) and the stove adoption benefit per subsidy dollar (defined as the benefit per stove adoption divided by the subsidy cost per stove adoption): $\text{MVPF} = 1 + \frac{\hat{v} + \hat{\Phi} - p}{\text{C}_s}$. This yields an MVPF of 3.1 when excluding environmental benefits ($\hat{\Phi} = 0$) and 13 when using a Social Cost of Carbon (SCC) of US\$120.

Figure B5: BDM and TIOLI elicitation results



Notes: Prior to the BDM each respondent completes two practice exercises, one for a bottle of lotion and one for a bar of soap. Each respondent is allocated a random price for the lotion and a random price for the soap. Respondents are randomly assigned to be offered the lotion via take-it-or-leave-it (TIOLI) and the soap via BDM, or vice versa. These two practice take-up decisions serve two purposes. First, participants learn how the BDM mechanism works relative to the TIOLI they are used to in stores and experience the binding nature of the BDM. Second, comparing the demand curves elicited through the TIOLI and BDM mechanisms provides a test of comprehension. Their similarity for both goods suggests that respondents understand the BDM mechanism and that the elicited WTP values reflect realistic decisions. The average price paid for these goods, US\$0.68, is small relative to the stove price and was deducted from the respondent's study compensation; it is therefore unlikely to affect short-term liquidity or WTP.

IV Online Appendix: Additional background information

IV.1 Policy background

Clean cooking policy in East Africa Given steady population growth, the IEA projects that approximately one billion people in Africa will still cook with biomass as their primary cooking fuel in 2050 (IEA, 2025). The rapid rise of household electricity access combined with the persistent use of biomass fuel for cooking has generated widespread policy interest in ‘leapfrogging’ populations towards electric cookstoves. According to the Organization of the Petroleum Exporting Countries (OPEC) Fund for International Development, “most specialists agree that electric cooking from renewable energy sources remains the ideal solution over the longer term” (OPEC, 2024).

To mitigate the negative externalities associated with charcoal production and to raise revenue for the financially distressed electric utility, in 2017 Kenya’s national electricity distributor launched “Pika na Power,” or “cook with electricity” (Kenya Power, 2017). It aims to increase uptake of electric cooking “to 5% (500,000 customers) in the short term and to 10% in the medium term” (Kenya Power, 2023). The Kenyan government’s “National Electric Cooking Strategy Action Plan” (Ministry of Energy, 2024b) announced VAT waivers for new electric appliances, behavior change campaigns, expanded support for consumer financing, investments in grid capacity and reliability, and an ‘e-cooking’ electricity tariff designed to make electric cooking more attractive relative to LPG and biomass stoves. This push follows Kenya’s national electrification program, which increased household grid access to over 58% in 2024 (KNBS, 2024).

Other countries are following suit: for example, Uganda’s 2022 Nationally Determined Contribution (NDC) was for “electricity to reach 50% of cooking fuel share by 2025” (Republic of Uganda, 2022).

Standards and bans Standards and bans are common alternatives to the direct pricing of externalities (Fowlie et al., 2020; Goulder and Parry, 2008), and can be more efficient than taxes when customers misperceive energy costs (Houde and Myers, 2025). Kenya has promoted the adoption of energy-efficient lighting through CFL distribution programs and standards (Kovacova, 2026; Wambui, 2010), and is now one of Africa’s largest LED markets (IMARC Group, 2025).

In the context of cooking, such policies face political and enforcement hurdles. Much of Kenya’s charcoal-related deforestation is informal, and national logging bans have been largely ineffective because of enforcement gaps (Sola and Cerutti, 2021; Daghar, 2021). Traditional cookstoves are often produced locally, and standards for them would face similar enforcement constraints. Given widespread informality in the charcoal and stove sectors, bans or standards likely require inefficiently costly enforcement.

IV.2 Credit policy

Study participants who bought the induction stove with a loan were subject to ECOA’s standard loan terms, which apply to all customers. Customers can choose their preferred payment frequency. In the lead-up to each payment deadline, customers who have not yet paid receive SMS reminders. Delinquent accounts are manually checked and followed up on by a dedicated agent team (BURN Manufacturing, 2023). If the customer does not make the required payment within seven days, the device is remotely turned off and only turned back on once the customer makes the payment (a ‘digital collateral’ practice that is standard in East Africa; see Gertler et al., 2024). If after 14 days the customer has still not made the required payment, the agent visits the customer in the field either to restructure the loan (if non-payment is due to unforeseen circumstances) or, if needed, to repossess the device.

IV.3 Willingness-to-pay

We measure WTP using the Becker et al. (1964) mechanism, building on the implementations developed in Berry et al. (2020), Dean (2024), and Berkouwer and Dean (2022). Neither the participant nor the enumerator knows the participant’s randomly assigned price. The enumerator conducts a binary search over the range of Ksh 1,500 to Ksh 18,000 (US\$12 to US\$139), first asking “If the price of the ECOA is Ksh 9,792 [US\$76], would you want to buy it?”, then proceeding to a higher or lower price based on the participant’s answer, and repeating this process until arriving at the nearest Ksh 10 (US\$0.08). After arriving at a final WTP, the participant opens the envelope. If and only if the price inside is less than or equal to their WTP, they buy the stove at that price. Since the price is fixed ex ante, the mechanism is incentive compatible. Since upfront prices are randomly assigned, purchasing is random conditional on WTP within the support of random prices. Practice BDM and take-it-or-leave-it (TIOLI) exercises result in similar estimates (Online Appendix Figure B5).

IV.4 Marginal-cost subsidy implementation

All participants have prepaid electricity meters, which are the norm in Kenya and across many African utilities. 87% of all Kenyan residential electricity customers have prepaid meters, and 95% of all residential customers who were connected to the grid in or after 2015—a group that includes most low-income connected households—do.²⁷ Consuming electricity runs down the meter: a customer must periodically ‘top up’ their meter to avoid its reaching zero, at which point electricity is shut off.²⁸ They can do so by purchasing a unique 20-digit token code either at a kiosk or via the Kenya Power mobile application, and then manually entering this code into their meter. We implement electricity transfers through this same channel: the research team uses the participant’s account number to purchase electricity tokens on their behalf, and then sends them the 20-digit token code via SMS messages. The participant then enters this manually into their meter. The code is unique to the meter and cannot be transferred, sold, or returned. We process electric stove usage data, calculate subsidy amounts, and send participants token codes on a weekly basis. For comparison, the median participant purchases electricity four times per month, or once every 7.5 days, suggesting our implementation matches participants’ own purchasing habits.

On the one hand, the manual entry of tokens could add an additional hurdle that lowers the value of marginal-cost subsidies. On the other hand, it could make the transfer more salient than an automatic bill rebate. Rather than trying to bound these opposing forces, we only note that the utility recently used this same method of token transfers to implement a subsidy pilot study.²⁹ Our methodology thus reflects how the electric utility would realistically implement such a program, so the empirical estimates can be interpreted as policy-relevant.

IV.5 Charcoal to CO₂e conversion

Intuitively, the molar mass of oxygen is approximately 16 g/mol and the mass of carbon (which charcoal largely consists of) is approximately 12 g/mol. Each carbon atom combines with two oxygen atoms to generate CO₂ such that 12 grams of charcoal generates 44 grams of CO₂, or 3.7 kg CO₂ per kg of charcoal. This stoichiometric calculation is a useful benchmark, and it is close to the factor we would obtain if all of the biomass were harvested renewably and no emissions arose outside

²⁷ Authors’ calculations from Kenya Power meter data.

²⁸ For 31% of our sample this had happened once or twice in the past month. For the remainder it had not.

²⁹ Kenya Power’s e-cooking tariff development study states that “Rebates are provided as electricity tokens sent via SMS to pre-paid customers” (Ministry of Energy, 2024a).

of combustion. However, it does not factor in production, processing, and transportation, which account for the majority of the total emissions factor; the purity of charcoal in East Africa (it does not consist exclusively of carbon); or regrowth rates, which determine what share of combustion emissions is offset by regrowth.

To obtain a more accurate estimate of emissions in our context, we convert quantities of charcoal into CO_{2e} using an emissions factor that combines combustion and non-combustion sources. Combustion emissions depend on how much of the wood used to produce the charcoal is harvested non-renewably, since wood that regrows reabsorbs the carbon released when it is burned. Estimates from Floess et al. (2023) imply combustion emissions of 86.8 tCO_{2e} per terajoule for the non-renewable fraction of biomass and 7.8 tCO_{2e} per terajoule for the renewable fraction. We assume a fraction of non-renewable biomass (fNRB) of 38%, following the most recent Modeling Fuelwood Savings Scenarios (MoFuSS) estimates for Kenya (Ghilardi and Bailis, 2024), which yields combustion emissions of 37.9 tCO_{2e} per terajoule.

The production, processing, and transportation of charcoal add a further 124.2 tCO_{2e} per terajoule of non-combustion emissions, computed from the non-combustion emissions factors of Floess et al. (2023) using the global warming potential for methane in biomass systems from Whitman and Lehmann (2011), following Gill-Wiehl et al. (2024). This generates a total emissions factor of 162 tCO_{2e} per terajoule. With a calorific value of 0.027 terajoules per ton of charcoal, this yields an estimated 4.4 kilograms of CO_{2e} per kilogram of charcoal.

The fNRB assumption is the most consequential input into this calculation, since it governs the weight placed on the two combustion factors. Holding all else fixed, an fNRB of 10% lowers the emissions factor to 3.8 kilograms of CO_{2e} per kilogram of charcoal, while an fNRB of 100% raises it to 5.7 kilograms.

Sensitivity to the fNRB assumption There is substantial uncertainty around the fNRB. Our primary estimates use the 38% rate that Ghilardi and Bailis (2024) estimate using their MoFuSS model, following the recommendations of the Article 6.4 mechanism of the Paris Agreement. To measure the sensitivity of our results to this parameter, we also calculate costs under 10% and 100% rates. This changes the subsidy cost per tCO_{2e} to US\$14 or US\$9, respectively (scenarios 3 and 4 of Appendix Table A3). These differences are relatively minor because most charcoal-related emissions are incurred not during combustion but during production and processing, which are not affected by the assumed fNRB (Floess et al., 2023).

Grid emissions intensity Kenya’s electricity generation mix is 85% renewable, generating less than half the emissions per kWh of the U.S. grid. However, even if Kenya’s grid were fully coal-powered, the subsidy cost per tCO_{2e} would remain essentially unchanged (scenario 5 of Appendix Table A3). Cooking with charcoal is far dirtier than cooking with electricity, so that the large reduction in charcoal cooking emissions is only very slightly offset by the increase in electricity cooking emissions (as discussed in Section 4.3).

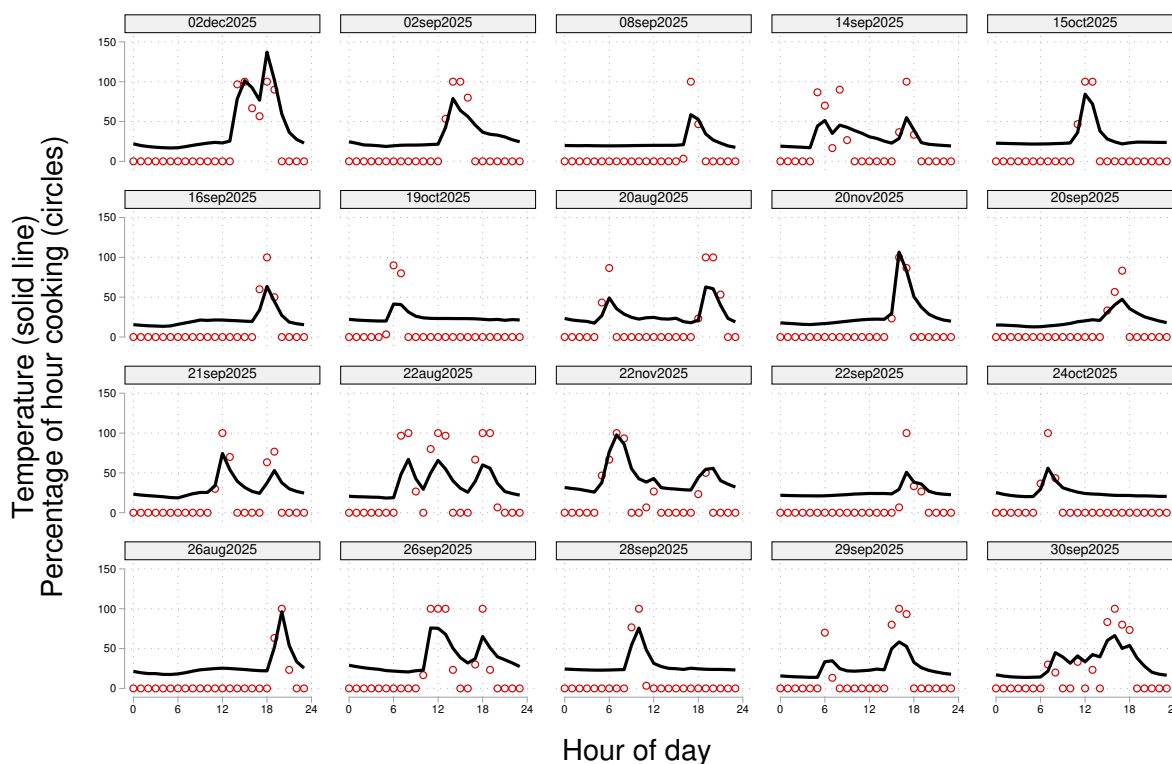
IV.6 Charcoal stove temperature sensors

Charcoal stove usage During the baseline visit, enumerators install temperature sensors to monitor usage of households’ charcoal stoves. We use the EXACT Pro sensor, a thermistor-based logger measuring 56×26×12 mm, weighing 20 g, sealed against water and dust, and rated for continuous operation up to 120°C (Climate Solutions, 2024). The device takes a measurement every minute and writes to internal memory at a user-configured interval, which we set to two minutes; its capacity of 130,000 logs corresponds to roughly six months at this interval, and its

battery lasts in excess of four years. The sensor measures the temperature of its stove-facing surface relative to ambient, so that the recorded series reflects heat from the stove rather than ambient warming. Data are downloaded several times using a handheld launcher that communicates with each sensor over Bluetooth Low Energy and writes to a microSD card. For households that own two or more charcoal stoves at baseline, we install monitoring devices on all stoves that are less than three years old (stoves older than this often cause the device to malfunction). The sensors are screwed into the stove body with an M4 self-tapping sheet-metal screw, which does not affect the stove’s functionality, and positioned at the back or outside of the stove so as not to obstruct the cook; placement is held consistent across households so that the temperature series remain comparable. [Figure 3](#) shows a charcoal stove instrumented with a temperature sensor. The installation occurs during the baseline visit so that we can estimate a within-individual treatment effect. Thomas et al. (2016) found no significant Hawthorne effects from electronic stove monitoring; still, any Hawthorne effect resulting from the installation of stove sensors would be symmetric across treatment groups.

We develop a basic algorithm to convert the two-minute temperature series to a two-minute cooking dataset, which we then aggregate to the hourly level for the analyses. [Figure B6](#) below shows several examples of participant-days of data.

Figure B6: Temperature-to-cooking conversion algorithm examples



Notes: Comparison of raw temperature data in Celsius (black line) and the estimated cooking fraction (red dots), for 20 randomly selected participant-days with at least one cooking event. Online Appendix IV.6 describes the technology and the conversion algorithm.

We restrict attention to participant-days with at least 90% of two-minute observations. For each participant-day we construct two reference points from the day’s own temperature distribution: a cold baseline, taken as the mean of the 15 coldest observations recorded between midnight and 6am,

and a hot reference, taken as the mean of the five hottest observations of the day. Using these, an observation is marked as ‘cooking’ if either of two conditions holds: the temperature exceeds 40°C, or the temperature has risen by at least 1°C in each of the three consecutive two-minute intervals ending at that observation. We then override this classification back to ‘not cooking’ when the stove is unambiguously cooling, defined as the temperature having fallen below the midpoint between that day’s cold baseline and hot reference while failing to rise by at least 1°C in any of the three preceding intervals. The first condition captures sustained high heat, the second captures the sharp temperature rise at the start of a cooking event, and the override prevents the long cooling tail of a charcoal fire from being counted as continued use.

For each participant we construct a complete two-minute grid running from the baseline visit through the final data download visit, so that periods without sensor coverage remain explicit rather than being silently dropped. Two further adjustments then draw on the survey data. First, for participants reporting at the endline survey that they no longer own a charcoal stove, all observations between the preceding visit and that visit are set to non-cooking. Second, we construct an imputed version of the series in which gaps in sensor coverage are set to non-cooking whenever the participant reported zero charcoal expenditure over the corresponding recall period; we use the imputed series as the main dataset. The two-minute indicators are then averaged within participant-hour to give the fraction of each hour spent cooking, which we scale to obtain hours and minutes of cooking per day, and which we split into the pre period (between the baseline and the main visit) and the post period (all observations thereafter).

We also test the sensitivity of our treatment effects to our choice of parameters in our algorithm. [Table B14](#) documents all the parameters at our disposal to incorporate in our temperature to cooking algorithm. We estimate treatment effects with inclusion of these other parameters that were not used in the main algorithm (min event, gradient, etc.). We also estimate treatment effects by using both conservative and relaxed mappings of temperature to cooking relative to the standard algorithm. [Tables B15](#) and [B16](#) show the estimates of this sensitivity analysis.

³⁰Gould et al. (2024) also uses the fixed cutoff of 40°C to define cooking events.

³¹Pillarisetti et al. (2014) also uses 40 min as the parameter choice to group up cooking events although their SUMS time series is at the 10 minute level.

Table B14: Parameters used in cooking event classification

Parameter	Description	Value
Rapid Rise Steps	Number of consecutive temperature increases of 1 degree Celsius needed to classify a reading as cooking	3
Temperature Threshold	Level of temperature that needs to be exceeded by the temperature sensor	40°C ³⁰
Cooldown	Once a cooking episode is flagged, cooking is considered over when temperature stops rising and falls below this point between the day’s cold baseline and hot reference	Midpoint
Smoothing	Controls how much each temperature observation is smoothed before checking it against threshold values; coefficient represents how much weight we give to the current temperature reading	0.9
Δ Temp	Change in temperature necessary to be classified as cooking	0.2°C/min
Gradient	Computes difference between the cookstove’s hot side and cool side; aims to rule out arbitrary jumps in ambient temperature	0.2°C
Min Event	A detected cooking event shorter than this number of minutes is considered noise and not cooking	3 min
Merge Event	Two cooking events that are close enough in time, as characterized by this measure, are combined together	40 min ³¹
Smoothing Reset Gap	If the sensor misses readings for longer than this, the smoothing resets	6 min

Notes: The main results use the first three parameters only in their cooking classification. Other tables test robustness by using the other parameters and also test sensitivity to parameter value choices by changing the values of these parameters in ways that unambiguously affect cooking time in a particular direction.

Table B15: Robustness of pre and post period stove usage effects to the cooking measure

	Temperature (raw)	Temperature (smoothed)			Gradient		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Main	Standard	Conservative	Relaxed	Standard	Conservative	Relaxed
Pre-period difference	-0.10 (0.17)	-0.12 (0.18)	-0.11 (0.14)	-0.14 (0.22)	-0.11 (0.09)	-0.08 (0.06)	-0.09 (0.12)
Post-period difference	-0.50*** (0.16)	-0.52*** (0.16)	-0.42*** (0.13)	-0.62*** (0.20)	-0.27*** (0.08)	-0.19*** (0.05)	-0.34*** (0.11)
Observations	223814	223814	223814	223814	223814	223814	223814
Respondents	2009	2009	2009	2009	2009	2009	2009
Control Mean	1.52	1.58	1.26	1.93	0.75	0.51	0.99
Level of data	Daily	Daily	Daily	Daily	Daily	Daily	Daily
Respondent FE?	No	No	No	No	No	No	No

Notes: Each column re-estimates column 5 of Appendix Table A2 in the main paper (the pre and post period specification), changing only the algorithm that converts the two-minute temperature data into cooking time. Column 1 uses the raw temperature data and the main classification rules of Table B14, and therefore matches the main specification in Appendix Table A2. Columns 2–4 use the smoothed series and add the Min Event and Merge Event parameters: column 2 at the values in Table B14; column 3 with a more conservative definition (a temperature cutoff of 45°C, a ramp-up of four periods, a cooldown at 60% of the way from the cold baseline to the hot reference, a minimum event length of five minutes, and a merge window of 30 minutes); column 4 with a more relaxed definition (35°C, two periods, 40%, three minutes, and 50 minutes). Columns 5–7 instead classify cooking using the Δ Temp and Gradient parameters: column 5 at the values in Table B14, column 6 at 0.3°C/min and 0.3°C, and column 7 at 0.1 and 0.1, with the same minimum event lengths and merge windows as columns 2, 3, and 4, respectively. All regressions cluster standard errors by participant. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B16: Robustness of difference-in-differences stove usage effects to the cooking measure

	Temperature (raw)	Temperature (smoothed)			Gradient		
	(1) Main	(2) Standard	(3) Conservative	(4) Relaxed	(5) Standard	(6) Conservative	(7) Relaxed
Pre-post difference	-0.30** (0.13)	-0.31** (0.13)	-0.25** (0.11)	-0.36** (0.16)	-0.13* (0.07)	-0.09* (0.05)	-0.18** (0.09)
Observations	223814	223814	223814	223814	223814	223814	223814
Respondents	2009	2009	2009	2009	2009	2009	2009
Control Mean	1.52	1.58	1.26	1.93	0.75	0.51	0.99
Level of data	Daily	Daily	Daily	Daily	Daily	Daily	Daily
Respondent FE?	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Each column re-estimates column 6 of Appendix Table A2 in the main paper (the difference-in-differences specification), changing only the algorithm that converts the two-minute temperature data into cooking time. Column 1 uses the raw temperature data and the main classification rules of Table B14, and therefore matches the main specification in Appendix Table A2. Columns 2–4 use the smoothed series and add the Min Event and Merge Event parameters: column 2 at the values in Table B14; column 3 with a more conservative definition (a temperature cutoff of 45°C, a ramp-up of four periods, a cooldown at 60% of the way from the cold baseline to the hot reference, a minimum event length of five minutes, and a merge window of 30 minutes); column 4 with a more relaxed definition (35°C, two periods, 40%, three minutes, and 50 minutes). Columns 5–7 instead classify cooking using the Δ Temp and Gradient parameters: column 5 at the values in Table B14, column 6 at 0.3°C/min and 0.3°C, and column 7 at 0.1 and 0.1, with the same minimum event lengths and merge windows as columns 2, 3, and 4, respectively. All regressions cluster standard errors by participant. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

V Online Appendix: Pre-analysis plan analysis

A pre-analysis plan, [available here](#), was filed with the AEA RCT Registry, [ID: 16832](#) (Berkouwer and Dean, 2025). This section reports each outcome listed under *Primary hypotheses for the environmental economics of induction stove adoption* in the pre-analysis plan, and points to where it appears in the paper. All estimates use the instrumental variables strategy the plan specified, instrumenting adoption with the randomly assigned price and credit type. One deviation is that we add the interaction of the price and credit treatments, which is also randomly assigned, as a third instrument for additional first-stage power. The plan also specified a minimum household size of two; this was raised to three before launch to reflect usage patterns on the ground.

V.1 Preliminary outcomes

Minutes spent cooking with charcoal per day Charcoal cooking falls by 30 minutes per day, a 33% reduction against the complier control mean. Reported in column 1 of Panel A of [Table 2](#) and discussed in [Section 4.1](#). [Figure 4](#) shows the event study from four weeks before to twelve weeks after the main visit; Appendix [Table A2](#) reports the specification alongside alternatives at the hourly and daily level, with and without participant fixed effects.

Electricity expenditures and corresponding CO_{2e} emissions Monthly electricity spending rises by US\$2.5 (70%), in column 5 of Panel A of [Table 2](#); column 6 reports the same among the subsidy control group. The corresponding emissions increase is in column 3 of Panel B. Discussed in [Section 4.2](#) and [Section 4.3](#).

Charcoal expenditures and corresponding CO_{2e} emissions Monthly charcoal spending falls by US\$9.7 (64%), in column 2 of Panel A of [Table 2](#). The corresponding emissions reduction is in column 1 of Panel B.

LPG expenditures and corresponding CO_{2e} emissions Reported in column 3 of Panel A of [Table 2](#) and column 2 of Panel B. The point estimates are small and not statistically distinguishable from zero.

Aggregate energy expenditures and corresponding CO_{2e} emissions Total monthly energy spending falls by US\$8.1 (32%), in column 7 of Panel A of [Table 2](#), equivalent to US\$189 discounted over the stove’s 2.2-year expected life. Total emissions fall by 2.4 tCO_{2e} per year, in column 4 of Panel B, or 5.3 tCO_{2e} over the stove’s life.

V.2 Primary outcomes: cost per ton of CO_{2e} abated

Fixed-cost subsidies, credit control group US\$12 per tCO_{2e}. Reported in [Section 5.1](#) and [Table 6](#), with the sensitivity of the estimate to alternative assumptions in Appendix [Table A3](#).

Fixed-cost subsidies, credit treatment group US\$20 per tCO_{2e}, in the same table. The gap between the two is the paper’s central result and is decomposed into its two channels in [Table 6](#).

Usage subsidies, credit treatment and credit control groups The pre-analysis plan anticipated a cost per ton for each group. We do not report one, because the marginal-cost subsidies moved neither margin: the effect on willingness-to-pay is a precise zero, and we can rule out an effect larger than US\$0.2 per US\$1 of subsidy (column 3 of Appendix [Table A5](#)); the usage elasticity is -0.07, ruling out a decrease larger than 0.3% in response to a 1% price increase (column 7). A cost per ton divides subsidy expenditure by the abatement that expenditure induced. Here the denominator is indistinguishable from zero, so the ratio is not informatively bounded rather than large, and the informative statement is the bound on each margin rather than a ratio of one bound to another. We therefore report the two margins separately, in [Section 5.2](#) and [Figure 5](#).

This is a property of the results, not of the implementation. The subsidies were delivered as designed, through the same channel households use to buy electricity and at the same weekly frequency, and take-up of the transfers was not the constraint. The quantity the plan asked for is undefined because electric cooking did not respond to the price of electricity. For the same reason, we are also unable to separate the selection and treatment channels in the effect of the announced subsidy on emissions.

V.3 Determinants of aggregate subsidy efficacy

Marginal adopters incentivized per subsidy dollar Reported in [Table 6](#) as subsidy expenditure required per additional adoption, which includes payments to inframarginal adopters: US\$67 for those paying with cash and US\$96 for those paying with credit.

Externality avoided by each marginal adopter 2.6 tCO_{2e} per year among people paying with cash and 2.2 tCO_{2e} among people paying with credit, or 5.7 and 4.8 over the stove’s life. Reported in [Table 4](#) and [Table 6](#). The four proxies the plan named are reported separately: charcoal stove usage in [Figure 4](#) and column 1 of [Table 2](#), induction stove usage in Panel B of [Figure 5](#), and charcoal and electricity emissions in columns 1 and 3 of Panel B of [Table 2](#).

V.4 Secondary outcomes

Time savings None detected. Charcoal cooking falls by 29 minutes per day while induction cooking adds 35 minutes, leaving participants cooking 4.6 minutes more in total, with both stoves in use simultaneously for only 0.7% of stove time. Reported in columns 7–9 of Appendix [Table A2](#) and discussed in [Section 4.4](#).

Self-reported health Reductions in self-reported cough and breathlessness among participants and their children. Reported in Berkouwer and Dean (2026) rather than here, as part of that paper’s treatment of the participant-level impacts of adoption.

Whether WTP and induction stove usage vary by underlying power quality Reported in Berkouwer and Dean (2026), which measures outages and voltage at the customer’s own outlet and treats grid reliability as a moderator of adoption and usage.

VI Proofs

Notation follows [Section 2](#). Undistorted willingness-to-pay v has distribution F , density f and survival function $Q = 1 - F$. The distortion D is continuously differentiable with $D' > 0$. At net price $P = p - s_1$ the distorted marginal adopter is $\tilde{v} = D^{-1}(P)$, distorted demand is $\tilde{Q}(P) = Q(D^{-1}(P))$, and

$$\tilde{Q}'(P) = -\frac{f(\tilde{v})}{D'(\tilde{v})}, \quad \kappa(P) \equiv \frac{Q(P)}{-Q'(P)}, \quad \tilde{\kappa}(P) \equiv \frac{\tilde{Q}(P)}{-\tilde{Q}'(P)}, \quad h(v) \equiv \frac{f(v)}{Q(v)}. \quad (15)$$

All cutoffs are interior with positive density and positive remaining demand, and every function is differentiable where a derivative is taken, so integrals with variable limits are differentiated by the Leibniz rule. First-dollar statements are evaluated at $s_1 = s_2 = 0$.

VI.1 Return to a purchase price subsidy

This subsection derives the resource identity [\(1\)](#), the marginal return [\(2\)](#) and the average return [\(3\)](#) of [Section 2.1.3](#).

Resource identity. With $R(s_1) = \rho(p - s_1) \tilde{Q}(p - s_1)$ and $dP/ds_1 = -1$, $dR/ds_1 = -\rho'(P) \tilde{Q}(P) - \rho(P) \tilde{Q}'(P)$. Induced adoptions satisfy $d\tilde{Q}(p - s_1)/ds_1 = -\tilde{Q}'(P)$. Dividing and using [\(15\)](#),

$$\dot{R}(s_1) = \frac{dR/ds_1}{d\tilde{Q}/ds_1} = \frac{-\rho'(P) \tilde{Q}(P) - \rho(P) \tilde{Q}'(P)}{-\tilde{Q}'(P)} = \rho(p - s_1) - \rho'(p - s_1) \tilde{\kappa}(p - s_1),$$

which is [\(1\)](#).

Marginal return. With only the purchase subsidy in place,

$$W(s_1) = \int_{D^{-1}(p-s_1)}^{\infty} \mathcal{V}(v) dF(v) - R(s_1), \quad C(s_1) = s_1 \tilde{Q}(p - s_1).$$

Differentiating the lower limit and using [\(15\)](#),

$$\frac{dW}{ds_1} = \mathcal{V}(\tilde{v}) \frac{f(\tilde{v})}{D'(\tilde{v})} - \frac{dR}{ds_1} = -\tilde{Q}'(P) [\mathcal{V}(\tilde{v}) - \dot{R}(s_1)], \quad \frac{dC}{ds_1} = \tilde{Q}(P) - s_1 \tilde{Q}'(P) = -\tilde{Q}'(P) [s_1 + \tilde{\kappa}(P)].$$

Dividing,

$$\frac{dW/ds_1}{dC/ds_1} = \frac{\mathcal{V}(\tilde{v}) - \dot{R}(s_1)}{s_1 + \tilde{\kappa}(P)},$$

which is [\(2\)](#).

Average return. For $0 < s_1 < p$ with a positive mass of induced adopters, define

$$\Delta Q \equiv \tilde{Q}(p - s_1) - \tilde{Q}(p) > 0, \quad \zeta \equiv \frac{\tilde{Q}(p - s_1)}{\Delta Q}, \quad (16)$$

$$\bar{\mathcal{V}} \equiv \frac{1}{\Delta Q} \int_{D^{-1}(p-s_1)}^{D^{-1}(p)} \mathcal{V}(v) dF(v), \quad \Delta R \equiv R(s_1) - R(0). \quad (17)$$

The induced adopters are the types $D^{-1}(p - s_1) \leq v < D^{-1}(p)$, so $\bar{\mathcal{V}}$ is the average value of an induced adoption. Total expenditure is $\Delta C = s_1 \tilde{Q}(p - s_1) = s_1 \zeta \Delta Q$, and the change in the objective is

$$\Delta W = \int_{D^{-1}(p-s_1)}^{D^{-1}(p)} \mathcal{V}(v) dF(v) - \Delta R = \bar{\mathcal{V}} \Delta Q - \Delta R.$$

Dividing,

$$\frac{\Delta W}{\Delta C} = \frac{\bar{\mathcal{V}} - \Delta R / \Delta Q}{s_1 \zeta},$$

which is (3).

VI.2 Return to a usage subsidy

This subsection derives the adoption responses (4), the returns (6) and (7), and the two comparisons stated in words after (7) in Section 2.1.4: the purchase price subsidy induces weakly more adoptions per dollar, and the usage subsidy generates a higher return if and only if its intensive-margin advantage, net of the difference in resource cost, exceeds the purchase price subsidy's extensive-margin advantage.

Setup. Let $b(v, s_2)$ denote the increase in type v 's private value of adoption generated by the usage subsidy, with $b(v, 0) = 0$ for all v . Usage conditional on adoption is $x(v, s_2)$ with $x(v, 0) = x(v)$ and $x'(v) > 0$. Because usage is chosen optimally and the subsidy is paid per unit, the envelope theorem gives $b_2(v, 0) \equiv \partial b(v, 0) / \partial s_2 = x(v)$. The principal's value from adoption by type v is $\mathcal{V}(v, s_2)$, with $\mathcal{V}(v) \equiv \mathcal{V}(v, 0)$ and $\mathcal{V}_2 \equiv \partial \mathcal{V} / \partial s_2$. Type v adopts when $D(v + b(v, s_2)) \geq p - s_1$, so the cutoff $\tilde{v}(s_1, s_2)$ solves

$$\tilde{v} + b(\tilde{v}, s_2) = D^{-1}(p - s_1), \quad (18)$$

and adoption is $\tilde{Q}(s_1, s_2) = Q(\tilde{v}(s_1, s_2))$, with $\tilde{Q}(0, 0) = \tilde{Q}(p)$. The objective and expenditure are

$$W(s_1, s_2) = \int_{\tilde{v}(s_1, s_2)}^{\infty} \mathcal{V}(v, s_2) dF(v) - R(s_1, s_2), \quad (19)$$

$$C(s_1, s_2) = s_1 Q(\tilde{v}(s_1, s_2)) + s_2 \int_{\tilde{v}(s_1, s_2)}^{\infty} x(v, s_2) dF(v), \quad (20)$$

with $R(s_1, 0) = R(s_1)$ and $R_i \equiv \partial R(0, 0) / \partial s_i$. Average usage among adopters at the first dollar is $\bar{x} \equiv \tilde{Q}(p)^{-1} \int_{\tilde{v}}^{\infty} x(v) dF(v)$.

Lemma A1 (Cutoff and adoption responses). *At $s_1 = s_2 = 0$,*

$$\frac{\partial \tilde{v}}{\partial s_1} = -\frac{1}{D'(\tilde{v})}, \quad \frac{\partial \tilde{v}}{\partial s_2} = -x(\tilde{v}), \quad \frac{\partial \tilde{Q}}{\partial s_1} = \frac{f(\tilde{v})}{D'(\tilde{v})}, \quad \frac{\partial \tilde{Q}}{\partial s_2} = f(\tilde{v}) x(\tilde{v}).$$

Proof. Since $b(v, 0) = 0$ for all v , $b_v(v, 0) \equiv \partial b(v, 0) / \partial v = 0$. Differentiating (18) with respect to s_1 and using $(D^{-1})'(P) = 1/D'(D^{-1}(P))$,

$$(1 + b_v) \frac{\partial \tilde{v}}{\partial s_1} = -\frac{1}{D'(D^{-1}(p - s_1))},$$

which at $s_1 = s_2 = 0$ is the first claim. Differentiating (18) with respect to s_2 ,

$$(1 + b_v) \frac{\partial \tilde{v}}{\partial s_2} + b_2(\tilde{v}, s_2) = 0,$$

which at $s_1 = s_2 = 0$ gives $\partial \tilde{v} / \partial s_2 = -b_2(\tilde{v}, 0) = -x(\tilde{v})$. The adoption responses follow from $\partial \tilde{Q} / \partial s_i = -f(\tilde{v}) \partial \tilde{v} / \partial s_i$. $\square$

Lemma A2 (Marginal expenditure). *At $s_1 = s_2 = 0$, $\partial C / \partial s_1 = \tilde{Q}(p)$ and $\partial C / \partial s_2 = \tilde{Q}(p) \bar{x}$.*

Proof. Differentiate (20). Every term multiplied by s_1 or s_2 vanishes at $s_1 = s_2 = 0$, leaving $\partial C / \partial s_1 = Q(\tilde{v}) = \tilde{Q}(p)$ and $\partial C / \partial s_2 = \int_{\tilde{v}}^{\infty} x(v, 0) dF(v) = \tilde{Q}(p) \bar{x}$. $\square$

Proof of (4). Dividing the adoption responses in Lemma A1 by the marginal expenditures in Lemma A2 and using $\tilde{Q}(p) = Q(\tilde{v})$,

$$\left. \frac{d\tilde{Q}/ds_1}{dC/ds_1} \right|_{s_1=s_2=0} = \frac{1}{D'(\tilde{v})} \frac{f(\tilde{v})}{Q(\tilde{v})}, \quad \left. \frac{d\tilde{Q}/ds_2}{dC/ds_2} \right|_{s_1=s_2=0} = \frac{x(\tilde{v})}{\bar{x}} \frac{f(\tilde{v})}{Q(\tilde{v})},$$

which is (4). $\square$

Proof of (6). Differentiating (19) with respect to s_2 by the Leibniz rule and substituting Lemma A1,

$$\frac{\partial W}{\partial s_2} = -\mathcal{V}(\tilde{v}, 0) f(\tilde{v}) \frac{\partial \tilde{v}}{\partial s_2} + \int_{\tilde{v}}^{\infty} \mathcal{V}_2(v, 0) dF(v) - R_2 = \mathcal{V}(\tilde{v}) f(\tilde{v}) x(\tilde{v}) + \int_{\tilde{v}}^{\infty} \mathcal{V}_2(v, 0) dF(v) - R_2.$$

Dividing by $\partial C / \partial s_2 = \tilde{Q}(p) \bar{x}$ from Lemma A2, using $\tilde{Q}(p) = Q(\tilde{v})$ in the first term and the definition (5) of $\mathcal{I}_2$ in the second,

$$\left. \frac{dW/ds_2}{dC/ds_2} \right|_{s_1=s_2=0} = \frac{x(\tilde{v})}{\bar{x}} \frac{f(\tilde{v})}{Q(\tilde{v})} \mathcal{V}(\tilde{v}) + \mathcal{I}_2 - \frac{R_2}{\tilde{Q}(p) \bar{x}},$$

which is (6). $\square$

Proof of (7). Differentiating (19) with respect to s_1 and substituting Lemma A1,

$$\frac{\partial W}{\partial s_1} = -\mathcal{V}(\tilde{v}) f(\tilde{v}) \frac{\partial \tilde{v}}{\partial s_1} - R_1 = \frac{\mathcal{V}(\tilde{v}) f(\tilde{v})}{D'(\tilde{v})} - R_1.$$

Dividing by $\partial C / \partial s_1 = \tilde{Q}(p) = Q(\tilde{v})$,

$$\left. \frac{dW/ds_1}{dC/ds_1} \right|_{s_1=s_2=0} = \frac{1}{D'(\tilde{v})} \frac{f(\tilde{v})}{Q(\tilde{v})} \mathcal{V}(\tilde{v}) - \frac{R_1}{\tilde{Q}(p)},$$

which is (7).

The same expression is (2) at $s_1 = 0$, as stated in Section 2.1.4. At $s_1 = 0$ the denominator of (2) is $\tilde{\kappa}(p) = \tilde{Q}(p) D'(\tilde{v}) / f(\tilde{v})$ by (15), and the resource term is $\dot{R}(0) = R_1 / (-\tilde{Q}'(p)) = R_1 D'(\tilde{v}) / f(\tilde{v})$. Substituting both into $[\mathcal{V}(\tilde{v}) - \dot{R}(0)] / \tilde{\kappa}(p)$ gives the display above. $\square$

Comparison of the two instruments. The usage subsidy generates a larger first-dollar return than the purchase price subsidy if and only if

$$\mathcal{I}_2 - \left[\frac{R_2}{\tilde{Q}(p) \bar{x}} - \frac{R_1}{\tilde{Q}(p)} \right] > \left[\frac{1}{D'(\tilde{v})} - \frac{x(\tilde{v})}{\bar{x}} \right] \frac{f(\tilde{v})}{Q(\tilde{v})} \mathcal{V}(\tilde{v}). \quad (21)$$

Proof of (21). Subtracting (7) from (6),

$$\left. \frac{dW/ds_2}{dC/ds_2} \right|_0 - \left. \frac{dW/ds_1}{dC/ds_1} \right|_0 = \mathcal{I}_2 - \left[\frac{R_2}{\tilde{Q}(p)\bar{x}} - \frac{R_1}{\tilde{Q}(p)} \right] - \left[\frac{1}{D'(\tilde{v})} - \frac{x(\tilde{v})}{\bar{x}} \right] \frac{f(\tilde{v})}{Q(\tilde{v})} \mathcal{V}(\tilde{v}),$$

which is positive exactly when (21) holds. The first bracket is the difference in resource cost per dollar between the two instruments, $R_2/(\tilde{Q}(p)\bar{x}) - R_1/\tilde{Q}(p)$. By (4), the second bracket times $f(\tilde{v})/Q(\tilde{v})$ is the difference in adoptions induced per dollar, and $\mathcal{V}(\tilde{v})$ is the value of the marginal adopter, the same type under both instruments; their product is the purchase price subsidy's extensive-margin advantage. Hence the usage subsidy generates a higher return if and only if its intensive-margin advantage $\mathcal{I}_2$, net of the difference in resource cost, exceeds the purchase price subsidy's extensive-margin advantage, which is the comparison stated after (7).

The second bracket is weakly positive when $D'(\tilde{v}) \leq 1$. Since $x'(v) > 0$, $x(v) \geq x(\tilde{v})$ for every adopter $v \geq \tilde{v}$; integrating over adopters gives $\bar{x} \geq x(\tilde{v})$, so $x(\tilde{v})/\bar{x} \leq 1$. If $D'(\tilde{v}) \leq 1$ then $1/D'(\tilde{v}) \geq 1$, and the bracket is weakly positive. By (4) the purchase price subsidy induces weakly more adoptions per dollar, which is the statement after (7). $\square$

VI.3 Credit distortion

This subsection derives the statements of Section 2.2 that specialize the framework to $D(v) = v/(1+r)$ with resource cost $\rho(P) = rP$: the elasticity claim, the average return (8) and its positivity condition, the decomposition (9) and the signs of its selection and elasticity terms, the adoption responses (10), and the instrument comparison (11). Under this distortion $1/D' = 1+r$, $\tilde{v} = (1+r)P$, and $\tilde{Q}(P) = Q((1+r)P)$.

Elasticity. Since $\tilde{Q}'(P) = (1+r)Q'((1+r)P)$,

$$|\tilde{\mathcal{E}}(P)| = -\frac{P\tilde{Q}'(P)}{\tilde{Q}(P)} = -\frac{(1+r)PQ'((1+r)P)}{Q((1+r)P)} = |\mathcal{E}((1+r)P)|.$$

Distorted elasticity at P equals undistorted elasticity at $(1+r)P > P$, so $|\tilde{\mathcal{E}}(P)| \geq |\mathcal{E}(P)|$ whenever $|\mathcal{E}|$ is increasing in price. Because $|\mathcal{E}(v)| = v h(v)$ is the product of two positive increasing functions when h is increasing, an increasing hazard rate is sufficient. This is the statement in Section 2.2 that the distortion raises elasticity whenever $|\mathcal{E}|$ is increasing in price, with an increasing hazard rate f/Q sufficient.

Lemma A3 (Induced types under the credit distortion). *Fix $0 < s_1 < p$ and let $\underline{u}(r) \equiv (1+r)(p-s_1)$ and $\bar{u}(r) \equiv (1+r)p$, so that the induced adopters are the types $\underline{u}(r) \leq v < \bar{u}(r)$.*

- (i) $\zeta(r) = Q(\underline{u})/[Q(\underline{u}) - Q(\bar{u})]$ and $\zeta(r) - 1 = \tilde{Q}(p)/\Delta Q$. If h is increasing, $\zeta'(r) \leq 0$.
- (ii) The distribution of v conditional on $\underline{u}(r) \leq v < \bar{u}(r)$ increases in r in the sense of first-order stochastic dominance. Hence the average of any increasing function of v over induced adopters is increasing in r , and the average of any decreasing function is decreasing in r .

Proof. (i) By (17), $\Delta Q = Q(\underline{u}) - Q(\bar{u})$ and $\zeta = Q(\underline{u})/\Delta Q$, so $\zeta - 1 = Q(\bar{u})/\Delta Q = \tilde{Q}(p)/\Delta Q$. Write $\zeta = [1 - Q(\bar{u})/Q(\underline{u})]^{-1}$ with $Q(\bar{u})/Q(\underline{u}) = \exp(-\int_{\underline{u}}^{\bar{u}} h(v) dv)$. Differentiating the exponent with respect to r ,

$$\frac{d}{dr} \int_{\underline{u}(r)}^{\bar{u}(r)} h(v) dv = p h(\bar{u}) - (p - s_1) h(\underline{u}) \geq p h(\underline{u}) - (p - s_1) h(\underline{u}) = s_1 h(\underline{u}) > 0,$$

where the inequality uses $h(\bar{u}) \geq h(\underline{u})$ for $\bar{u} > \underline{u}$. The ratio $Q(\bar{u})/Q(\underline{u})$ is therefore decreasing in r , and ζ is decreasing in r .

(ii) The conditional distribution is $G_{\underline{u}, \bar{u}}(v) = [F(v) - F(\underline{u})]/[F(\bar{u}) - F(\underline{u})]$ for $\underline{u} \leq v < \bar{u}$, 0 below $\underline{u}$ and 1 from $\bar{u}$ on. For $\underline{u} \leq v < \bar{u}$,

$$\frac{\partial G_{\underline{u}, \bar{u}}(v)}{\partial \underline{u}} = \frac{f(\underline{u})[F(v) - F(\bar{u})]}{[F(\bar{u}) - F(\underline{u})]^2} \leq 0, \quad \frac{\partial G_{\underline{u}, \bar{u}}(v)}{\partial \bar{u}} = -\frac{f(\bar{u})[F(v) - F(\underline{u})]}{[F(\bar{u}) - F(\underline{u})]^2} \leq 0.$$

Raising $\underline{u}$ or $\bar{u}$ therefore weakly lowers $G_{\underline{u}, \bar{u}}(v)$ at every v inside the window; below the new $\underline{u}$ the distribution function is 0, and from the old $\bar{u}$ on the old distribution function is 1, so the inequality $G_{\underline{u}', \bar{u}'} \leq G_{\underline{u}, \bar{u}}$ holds at every v when $\underline{u}' \geq \underline{u}$ and $\bar{u}' \geq \bar{u}$. Both $\underline{u}(r)$ and $\bar{u}(r)$ are increasing in r , which gives the dominance ordering. The statements about averages follow from the definition of first-order stochastic dominance. $\square$

Average return. With $\rho(P) = rP$, $R(s_1) = r(p - s_1)\tilde{Q}(p - s_1)$. Using $\tilde{Q}(p - s_1) = \Delta Q + \tilde{Q}(p)$,

$$\Delta R = r(p - s_1)\tilde{Q}(p - s_1) - rp\tilde{Q}(p) = r(p - s_1)\Delta Q - rs_1\tilde{Q}(p),$$

which is the resource accounting stated in [Section 2.2.1](#): the subsidy generates $r(p - s_1)\Delta Q$ in resource cost among induced adoptions and saves $rs_1\tilde{Q}(p)$ among inframarginal adopters. By [Lemma A3\(i\)](#), $\tilde{Q}(p) = (\zeta - 1)\Delta Q$, so $\Delta R/\Delta Q = r(p - s_1) - rs_1(\zeta - 1) = rp - rs_1\zeta$. Substituting into (3),

$$\mathcal{W}(r) = \frac{\bar{\mathcal{V}}(r) - rp + rs_1\zeta(r)}{s_1\zeta(r)} = \frac{\bar{\mathcal{V}}(r) - rp}{s_1\zeta(r)} + r, \quad (22)$$

which is (8). The return is positive if and only if the numerator of the first expression is positive, that is, $\bar{\mathcal{V}}(r) + rs_1\zeta(r) > rp$, which is the condition stated after (8).

Effect of the interest rate. Differentiating (22) with respect to r , with primes denoting derivatives in r ,

$$\mathcal{W}'(r) = \frac{\bar{\mathcal{V}}'(r) - p}{s_1\zeta(r)} - \frac{[\bar{\mathcal{V}}(r) - rp]\zeta'(r)}{s_1\zeta(r)^2} + 1 = \frac{1}{s_1\zeta(r)} \left[\bar{\mathcal{V}}'(r) - [\bar{\mathcal{V}}(r) - rp] \frac{\zeta'(r)}{\zeta(r)} + s_1(\zeta(r) - 1) - (p - s_1) \right],$$

where the second equality uses $s_1\zeta - p = s_1(\zeta - 1) - (p - s_1)$. This is (9). By [Lemma A3\(ii\)](#), $\bar{\mathcal{V}}'(r) \geq 0$ when $\mathcal{V}$ is increasing over the window of induced types: a larger distortion raises $\bar{\mathcal{V}}$ by moving the induced group toward higher-value agents, which is the selection-channel statement after (9). By [Lemma A3\(i\)](#), $\zeta'(r) \leq 0$ when h is increasing: an increase in r weakly lowers $\zeta(r)$ and, whenever $\bar{\mathcal{V}}(r) \geq rp$, the elasticity term in (9) is nonnegative, which is the elasticity-channel statement after (9).

Comparison of the two instruments. With $1/D' = 1 + r$ and $x'(v) > 0$, (4) and the inequality $x(\tilde{v})/\bar{x} \leq 1$ from the proof of (21) give

$$\left. \frac{d\tilde{Q}/ds_1}{dC/ds_1} \right|_{s_1=s_2=0} = (1 + r) \frac{f(\tilde{v})}{Q(\tilde{v})} \geq \frac{x(\tilde{v})}{\bar{x}} \frac{f(\tilde{v})}{Q(\tilde{v})} = \left. \frac{d\tilde{Q}/ds_2}{dC/ds_2} \right|_{s_1=s_2=0},$$

which is (10).

Under the credit distortion the amount financed is the net purchase price, so with both instruments in place $R(s_1, s_2) = r(p - s_1) \tilde{Q}(s_1, s_2)$. By [Lemma A1](#) with $1/D' = 1 + r$,

$$R_1 = -r\tilde{Q}(p) + rp(1 + r)f(\tilde{v}), \quad R_2 = rp f(\tilde{v}) x(\tilde{v}).$$

Each induced purchase incurs financing resource cost rp , and each purchase-subsidy dollar paid to existing adopters saves r ; the usage subsidy leaves the amount financed unchanged. Substituting into (7) and (6), and writing $h = f(\tilde{v})/Q(\tilde{v})$ with $\tilde{Q}(p) = Q(\tilde{v})$,

$$\left. \frac{dW/ds_1}{dC/ds_1} \right|_0 = (1 + r) h [\mathcal{V}(\tilde{v}) - rp] + r, \quad \left. \frac{dW/ds_2}{dC/ds_2} \right|_0 = \frac{x(\tilde{v})}{\bar{x}} h [\mathcal{V}(\tilde{v}) - rp] + \mathcal{I}_2.$$

The purchase price subsidy has the larger first-dollar return if and only if the first expression exceeds the second,

$$[\mathcal{V}(\tilde{v}) - rp] \left[(1 + r) - \frac{x(\tilde{v})}{\bar{x}} \right] \frac{f(\tilde{v})}{Q(\tilde{v})} + r > \mathcal{I}_2,$$

which is (11).

VII Model extensions: derivations

Each subsection derives the statement of the paragraph of the same name in [Section 2.3](#). Throughout, the credit distortion of [Section VI.3](#) applies, and first-dollar statements are evaluated at $s_1 = s_2 = 0$.

VII.1 Marginal cost of raising funds

Let each dollar of subsidy expenditure carry an excess burden $\lambda - 1 \geq 0$, so the objective is $W - (\lambda - 1)C$. Per induced adoption, a marginal purchase subsidy generates

$$\mathcal{V}(\tilde{v}) - \dot{R}(s_1) - (\lambda - 1)(s_1 + \tilde{\kappa}(P)).$$

The distortion enters the excess-burden term only through $\tilde{\kappa}$. Since $\tilde{\kappa}(P)/\kappa(P) = |\mathcal{E}(P)|/|\tilde{\mathcal{E}}(P)|$, and $|\tilde{\mathcal{E}}(P)| \geq |\mathcal{E}(P)|$ under an increasing hazard rate by [Section VI.3](#), $\tilde{\kappa}(P) \leq \kappa(P)$. Holding the value of the induced adoption fixed, the distorted market saves $(\lambda - 1)[\kappa(P) - \tilde{\kappa}(P)] \geq 0$ in excess burden per induced adoption relative to the undistorted market, a saving that is increasing in λ and zero at $\lambda = 1$. Hence, under the increasing-hazard-rate condition, a positive marginal cost of raising funds increases the relative marginal benefit of subsidies in a distorted market per induced adoption, because the distortion reduces inframarginal spending, which is the statement of the paragraph.

VII.2 Taxation

Let the usage margin be corrected by a Pigouvian tax, so that the social value of adoption by type v is $v - p$ and the first best admits exactly the types $v \geq p$.

Under a pure allocation distortion ($\rho = 0$) the agent adopts if and only if $D(v) \geq p - s_1$. With $s_1 = p - D(p)$ the agent adopts if and only if $D(v) \geq D(p)$, which holds if and only if $v \geq p$ because D is strictly increasing. The adoption set is the first-best set and no resources are consumed. Hence, under a pure allocation distortion, setting $s_1 = p - D(p)$ achieves the first-best allocation, which is the first statement of the paragraph.

Under the credit distortion the agent adopts if and only if $v \geq (1 + r)(p - s_1)$ and each purchase consumes $r(p - s_1)$. Aligning the adoption margin with the first best requires $(1 + r)(p - s_1) = p$, that is $s_1 = rp/(1 + r)$, and each purchase then still consumes $rp/(1 + r) > 0$. Removing the resource cost requires $s_1 = p$, at which the threshold is zero and every agent with $v \geq 0$ adopts; since $p > 0$, this admits every type with $0 \leq v < p$, whose adoption lowers welfare. Hence, under a resource distortion, no purchase price subsidy restores the first best: eliminating the resource cost requires $s_1 = p$, which induces over-adoption whenever F places positive mass on $[0, p)$, and any smaller subsidy leaves some resource cost in place, which is the second statement of the paragraph.

Welfare is

$$W(s_1) = \int_{(1+r)(p-s_1)}^{\infty} (v - p) dF(v) - r(p - s_1) Q((1 + r)(p - s_1)).$$

Differentiating, with $T \equiv (1 + r)(p - s_1)$ the adoption threshold,

$$W'(s_1) = (1 + r)[T - p]f(T) + rQ(T) - r(p - s_1)(1 + r)f(T) = rQ(T) - (1 + r)s_1 f(T).$$

The first term is the financing resource cost saved on every existing purchase. The second is the welfare loss from the marginal adopter: their net social value $v - p - r(p - s_1)$ evaluated at $v = T$ equals $-s_1$, which is negative for every $s_1 > 0$, times the $(1 + r)f(T)$ adoptions induced per subsidy

dollar. At $s_1 = 0$, $W'(0) = r\tilde{Q}(p) > 0$, so the optimal subsidy is strictly positive. Hence the planner chooses a second-best $s_1 > 0$ that trades off the welfare loss from over-adoption at the margin against the resource cost saved on every adoption, which is the third statement of the paragraph.

VII.3 Direct instruments

Suppose the principal can remove the allocation distortion directly for the marginal agents that a first-dollar purchase subsidy would induce, at economic cost c per induced adoption. At the first dollar, the purchase subsidy's outlay per induced adoption is $\tilde{\kappa}(p)$, all of it paid to inframarginal agents; its economic cost per induced adoption is zero if subsidy payments are pure transfers and positive if raising the funds carries a marginal cost of public funds. Both instruments induce the same agents, so the value of induced adopters is common to both, and under an allocation distortion neither changes resource use. The direct instrument is therefore preferred whenever c does not exceed the subsidy's economic cost per induced adoption, which is the statement of the paragraph. Correcting a resource distortion directly also removes resource cost on existing purchases; [Section VII.4](#) quantifies this.

VII.4 Correcting the resource distortion

Evaluate at $s_1 = 0$. A marginal purchase subsidy ds_1 lowers the amount financed by every existing adopter, saving $r\tilde{Q}(p) ds_1$ in financing resources, and induces $(1+r)f((1+r)p) ds_1$ adoptions. Its financing-resource saving per induced adoption is

$$\frac{r\tilde{Q}(p)}{(1+r)f((1+r)p)} = r\tilde{\kappa}(p),$$

using $\tilde{\kappa}(p) = Q((1+r)p)/[(1+r)f((1+r)p)]$ from [\(15\)](#). A marginal reduction dr lowers the threshold $(1+r)p$ by $p dr$, inducing $f((1+r)p)p dr$ adoptions, and lowers the financing cost rp of every existing purchase by $p dr$, saving $p\tilde{Q}(p) dr$. Its saving per induced adoption is

$$\frac{p\tilde{Q}(p)}{f((1+r)p)p} = (1+r)\tilde{\kappa}(p).$$

These are the two savings stated in the paragraph: a purchase price subsidy saves $r\tilde{\kappa}$ in financing resources per induced adoption, whereas reducing r saves $(1+r)\tilde{\kappa}$. Throughout, “cost per induced adoption” means the economic cost per induced adoption, before netting out financing-resource savings. This could be zero if both policies are fully transfers or both positive if correcting the distortion/paying the subsidy incur a cost of public funds. Both policies move the same threshold and induce the same marginal types, so the value $\mathcal{V}(\tilde{v})$ of an induced adoption is common to both and drops out of the comparison. The objective therefore ranks the two policies by economic cost net of financing-resource savings per induced adoption. At equal economic cost, correcting r generates an additional $\tilde{\kappa}(p)$ of resource savings and a larger return. More generally, correction is preferred whenever its economic cost per induced adoption exceeds the subsidy's by no more than $\tilde{\kappa}(p)$, so it can remain preferable even when its cost exceeds the subsidy's.

VII.5 Subsidizing the resource distortion

Let the principal pay $\varsigma(p - s_1)$ of each buyer's resource cost, so the buyer faces the rate $r - \varsigma$ and adopts if and only if $v \geq (1+r-\varsigma)(p - s_1)$. Evaluate both instruments at the first dollar, $s_1 = \varsigma = 0$,

where the threshold is $(1+r)p$ and adoption is $\tilde{Q}(p)$; at $s_1 > 0$ the marginal cost of s_1 would also include s_1 times the induced adoptions. A marginal $d\varsigma$ is paid on every purchase, costing $p\tilde{Q}(p)d\varsigma$, and lowers the adoption threshold by $p d\varsigma$: one dollar of threshold reduction per $\tilde{Q}(p)$ dollars spent. A marginal ds_1 costs $\tilde{Q}(p)ds_1$ and lowers the threshold by $(1+r)ds_1$: $1+r$ dollars of threshold reduction per $\tilde{Q}(p)$ dollars spent. Per dollar of expenditure the purchase subsidy therefore moves the adoption threshold $1+r$ times as far, because each dollar of s_1 removes a dollar of purchase cost and the resource cost on that dollar. Both instruments pay every buyer and operate on the same threshold, so the purchase price subsidy induces $1+r$ times as many adoptions per dollar, a ratio that increases in r . Hence the purchase price subsidy generates a larger extensive-margin return than subsidizing the resource cost, increasingly so as r rises, which is the statement of the paragraph.

VII.6 Marginal distortions

This subsection derives the three statements of the marginal-distortions paragraph in [Section 2.3](#): a marginal distortion lowers WTP at least proportionally; it raises the true private benefit of the marginal adopter, while its effect on the marginal adopter's usage depends on the distortion; and it raises the demand elasticity under an increasing hazard rate whenever $D' \leq 1$.

Setup. The agent values each unit of fuel savings at a fraction $\delta \in (0,1)$ of its true value and chooses usage against that perceived return. Let $B(x) \geq 0$ denote fuel savings at usage x , increasing in x , and $C(x) \geq 0$ the switching cost. True and perceived WTP are

$$v = \max_x [B(x) - C(x)], \quad D(v) = \max_x [\delta B(x) - C(x)].$$

Let $x(v)$ denote the usage that maximizes the first problem, assumed increasing in v , and $\tilde{x}(v)$ the usage that maximizes the second. Because the perceived marginal return to usage is lower, $\tilde{x}(v) \leq x(v)$: adopters underuse the technology.

WTP falls at least proportionally. For every x , $\delta B(x) - C(x) \leq \delta[B(x) - C(x)]$ because $C(x) \geq 0$ and $\delta < 1$. Taking the maximum of each side gives $D(v) \leq \delta v < v$ for $v > 0$.

Private benefit of the marginal adopter. The distorted marginal adopter is $\tilde{v} = D^{-1}(P)$. Since $D(P) < P$ and D is increasing, $\tilde{v} > P = \underline{v}$: the distortion unambiguously raises the true private benefit of the marginal adopter.

Usage of the marginal adopter. The change in the marginal adopter's usage decomposes as

$$\tilde{x}(\tilde{v}) - x(P) = \underbrace{\left[\tilde{x}(\tilde{v}) - x(\tilde{v}) \right]}_{\text{underuse} \leq 0} + \underbrace{\left[x(\tilde{v}) - x(P) \right]}_{\text{higher type} \geq 0}.$$

The first term is the underuse of a given type and the second is the higher type of the marginal adopter. Their relative size depends on how the distortion reshapes usage across types, and either can dominate. How the higher private benefit of the marginal adopter maps into usage therefore depends on the distortion.

Elasticity. Distorted demand is $\tilde{Q}(P) = Q(D^{-1}(P))$, so distorted elasticity is

$$|\tilde{\mathcal{E}}(P)| = -\frac{P\tilde{Q}'(P)}{\tilde{Q}(P)} = \frac{Ph(D^{-1}(P))}{D'(D^{-1}(P))}.$$

Undistorted elasticity is $|\mathcal{E}(P)| = Ph(P)$. Two differences determine which is larger: the hazard rate is evaluated at the higher type $D^{-1}(P) > P$, and the expression is divided by D' .

To sign the effect of dividing by D' , we assume $D' \leq 1$: a household whose true WTP is one dollar higher has perceived WTP at most one dollar higher. This holds if households differ only in the value of fuel savings ($B = ab(x)$, with a varying across households), only in a multiplicative scale on a switching cost that rises with usage ($C = c\gamma(x)$, $\gamma' \geq 0$, with c varying), or only in a fixed switching cost: by the envelope theorem, with x and $\tilde{x} \leq x$ the true and perceived usage of a given household, $D' = \delta b(\tilde{x})/b(x) \leq \delta$ in the first case, $D' = \gamma(\tilde{x})/\gamma(x) \leq 1$ in the second, and $D' = 1$ in the third. These are the common utility functions referred to in the main text.

Under this assumption, an increasing hazard rate is sufficient for the distortion to weakly raise elasticity: an increasing hazard rate gives $h(D^{-1}(P)) \geq h(P)$, and $D' \leq 1$ means dividing by D' does not lower the expression, so $|\tilde{\mathcal{E}}(P)| \geq |\mathcal{E}(P)|$.

The increasing hazard rate cannot in general be relaxed to the weaker condition used for the credit distortion. There D is proportional to v , and the weaker condition that elasticity is weakly increasing in price suffices (Section VI.3); an increasing hazard rate implies it, since $|\mathcal{E}(P)| = Ph(P)$. Here D is generally not proportional, and that weaker condition can fail. For example, let $D(v) = \delta v - k$ with $k > 0$; this is the first case above with usage held fixed at some $\bar{x}$, so that $k = (1 - \delta)C(\bar{x})$. Let undistorted elasticity be a constant $|\mathcal{E}|$, which satisfies the weaker condition while the hazard rate $h(v) = |\mathcal{E}|/v$ falls. Distorted elasticity is then $|\mathcal{E}|P/(P + k) < |\mathcal{E}|$.

VII.7 Negative correlation between private and social benefits

Suppose $\Phi(v)$ is decreasing over the relevant range. At the margin, $\tilde{v} \geq v$ gives $\Phi(\tilde{v}) \leq \Phi(v)$: the purchase price subsidy margin generates a weakly lower external benefit than in the undistorted market, which is the first statement of the paragraph. For a finite subsidy under the credit distortion, Lemma A3(ii) gives $\bar{\mathcal{V}}'(r) \leq 0$ whenever $\mathcal{V}$ is decreasing over the window of induced types, so the selection term in (9) is weakly negative. The resource terms do not depend on Φ and are unchanged. Because ζ does not depend on Φ either, $\zeta'(r) \leq 0$ still holds under an increasing hazard rate, so the elasticity term remains nonnegative as long as $\bar{\mathcal{V}}(r) \geq rp$; if $\bar{\mathcal{V}}(r) < rp$ it is nonpositive. Hence the elasticity channel can still favor purchase price subsidies and the sign of $\mathcal{W}'(r)$ is not determined by the theory, which is the second statement of the paragraph.

References

- Becker, G. M., M. H. DeGroot, and J. Marschak (1964). “Measuring utility by a single-response sequential method”. *Behavioral Science* 9.3.
- Berkouwer, S. B. and J. T. Dean (2022). “Credit, attention, and externalities in the adoption of energy efficient technologies by low-income households”. *American Economic Review* 112.10.
- (2025). *Pre-Analysis Plan: Induction Stove Adoption in Kenya*. Tech. rep. AEA RCT Registry ID: AEARCTR-0016832. AEA RCT Registry.
- (2026). “Watts for Supper? The socioeconomic drivers and impacts of electric stove adoption in Kenya”. Working Paper.
- Berry, J., G. Fischer, and R. P. Guiteras (2020). “Eliciting and Utilizing Willingness to Pay: Evidence from Field Trials in Northern Ghana”. *Journal of Political Economy* 128.4.
- BURN Manufacturing (2023). *Burn Credit Policy*. Unpublished internal policy document. Version 1. Dated 23 May 2023.
- Chernozhukov, V., M. Demirer, E. Duflo, and I. Fernández-Val (2025). “Fisher–Schultz Lecture: Generic Machine Learning Inference on Heterogeneous Treatment Effects in Randomized Experiments, With an Application to Immunization in India”. *Econometrica* 93.4.
- Climate Solutions (2024). *EXACT Pro Sensor User Manual*. Version dated 13 May 2024. Available at https://www.climate-solutions.net/images/EXACT_Pro_Manual_5_13_2024.pdf.
- Daghar, M. (2021). “Kenya’s charcoal bans have fuelled a smuggling problem”. *ISS Today*. Published under the ENACT project, funded by the European Union.
- Dean, J. T. (2024). “Noise, Cognitive Function, and Worker Productivity”. *American Economic Journal: Applied Economics* 16.4.
- Floess, E., A. Grieshop, E. Puzzolo, D. Pope, N. Leach, et al. (2023). “Scaling up gas and electric cooking in low- and middle-income countries: climate threat or mitigation strategy with co-benefits?” *Environmental Research Letters* 18.3.
- Fowlie, M., M. Reguant, and S. P. Ryan (2020). *In Defense of Alternatives to Pollution Pricing*. Policy Brief. Accessed: 2026-03-26. Economics for Inclusive Prosperity (EfIP).
- Gertler, P., B. Green, and C. Wolfram (2024). “Digital Collateral”. *The Quarterly Journal of Economics* 139.3.
- Ghilardi, A. and R. Bailis (2024). *Updated fNRB Values for Woodfuel Interventions*. Tech. rep.
- Gill-Wiehl, A., D. M. Kammen, and B. K. Haya (2024). “Pervasive over-crediting from cookstove offset methodologies”. *Nature Sustainability* 7.2.
- Gould, C. F., L. Davila, M. L. Bejarano, M. Burke, D. W. Jack, et al. (2024). “Exposure to Nitrogen Dioxide and Fine Particulate Matter When Cooking with Electricity Compared to Gas, a Randomized Crossover Study in Quito, Ecuador”. *Environmental Health Perspectives* 132.1.
- Goulder, L. H. and I. W. H. Parry (2008). “Instrument Choice in Environmental Policy”. *Review of Environmental Economics and Policy* 2.2.
- Hendren, N. and B. Sprung-Keyser (2020). “A Unified Welfare Analysis of Government Policies”. *The Quarterly Journal of Economics* 135.3.
- Houde, S. and E. Myers (2025). “Heterogeneous (Mis)Perceptions of Energy Costs: Implications for Measurement and Policy Design”. *Journal of Political Economy Microeconomics*.
- IMARC Group (2025). *East Africa LED Lighting Market: Industry Trends, Share, Size, Growth, Opportunity and Forecast 2025-2033*. Tech. rep.
- International Energy Agency (2025). *Universal Access to Clean Cooking in Africa: Progress Update and Roadmap to Implementation*. Accessed: 2026-03-26. Paris: International Energy Agency.
- Kenya National Bureau of Statistics (2024). *Kenya Housing Survey Basic Report*. Tech. rep.

- Kenya Power (2017). *Annual Report and Financial Statements for the Year Ended 30 June 2017*. Tech. rep.
- (2023). *Kenya Power Annual Report and Financial Statements for the Year Ended 30 June 2023*.
- Kovacova, M. (2026). *Energy efficiency in Kenya: Light bulbs*. illuminem.
- Ministry of Energy and Petroleum, Republic of Kenya (2024a). *Kenya National Electric Cooking Strategy (KNeCS)*. National Strategy Report. Accessed: 2026-01-28.
- (2024b). *Kenya National Electric Cooking Strategy: Action Plan*. Action Plan. Accessed: 2026-01-07. Kenya Power and Lightning Company Ltd.
- OPEC Fund for International Development (2024). *Clean Cooking Report 2024*. Knowledge Series Report. Accessed: 2026-01-07. OPEC Fund for International Development.
- Pillarisetti, A., M. Vaswani, D. Jack, K. Balakrishnan, M. N. Bates, et al. (2014). “Patterns of Stove Usage after Introduction of an Advanced Cookstove: The Long-Term Application of Household Sensors”. *Environmental Science and Technology* 48.24.
- Republic of Uganda (2022). *Updated Nationally Determined Contribution (NDC) — Republic of Uganda*. Nationally Determined Contribution. Submitted to the UNFCCC on 12 September 2022; Accessed: 2026-01-07.
- Sola, P. and P. O. Cerutti (2021). “Kenya has been trying to regulate the charcoal sector: why it’s not working”. *The Conversation*. Republished by World Agroforestry (ICRAF).
- Thomas, E. A., S. Tellez-Sanchez, C. Wick, M. Kirby, L. Zambrano, et al. (2016). “Behavioral Reactivity Associated With Electronic Monitoring of Environmental Health Interventions—A Cluster Randomized Trial with Water Filters and Cookstoves”. *Environmental Science and Technology* 50.7.
- Wambui, S. (2010). *Kenya launches energy saving initiative*. Capital FM Business.
- Whitman, T. and J. Lehmann (2011). “Systematic under-and overestimation of GHG reductions in renewable biomass systems: A Letter”. *Climatic Change* 104.2.